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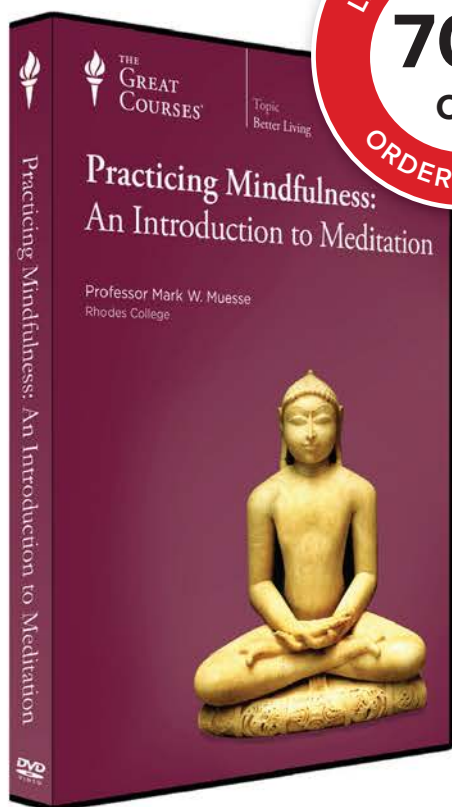
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38 Three children sit in front of a cellar door in San Francisco's Chinatown around the turn of the 20th century. New archaeological projects are examining the lives of America's first Chinese immigrants.

Cover: Traces of paint remain on a 14th-century B.C. relief depicting the pharaoh Akhenaten and his family offering lotus flowers to the god Aten, represented as a sun disc.

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Unexpected Narratives

More than 3,000 years ago, the pharaoh Akhenaten elected to move Egypt's capital from Thebes to an uninhabited spot along the Nile. He dedicated his magnificent new city, Amarna, to the god Aten. Archaeologist Barry Kemp, who has devoted his long career to the study of Amarna, recalls his own response to a stretch of desert there: "The surface was relatively smooth and undulating, indicative of the presence of small houses." Contributing editor Heather Pringle brings us "Beyond the Palace Walls" (page 24), the story of the search not just for palatial remains, but for subtle evidence of the lives



of the common people who constructed Amarna's palaces and temples.

Much of what we have known until now about the Comanche has come from the work of historians. While the Comanche were known as superb horsemen and audacious raiders, only recently has archaeological research been brought to bear on understanding the nature of their culture and their true role in the expansion of the western United States. In "Searching for the

Comanche Empire" (page 44), online editor Eric A. Powell brings us word of discoveries that are dramatically expanding our knowledge of this Native American nation.

A variety of circumstances conspired to create the surge of Chinese immigrants to America in the mid-1800s. In "America's Chinatowns" (page 38), deputy editor Samir S. Patel surveys the most recent archaeological research into the Chinese immigrant experience. An examination of the things they used day-to-day both in cities and in work camps is allowing archaeologists to construct a rich picture of what their lives were really like.

"Letter from Philadelphia: City Garden" (page 53), by journalist Margaret Shakespeare, covers what can be thought of as a rare category of archaeological site—a relatively untouched, bucolic spread of land in the middle of an American city. Evidence of occupation at Bartram's Garden goes back as far as 4,500 years. Artifacts there are demonstrating that prehistoric peoples used the site seasonally to collect wild plants and to fish the waters of what is now the Schuylkill River. Also present is later evidence of successive occupations by a French Huguenot, the Bartram family, and a railroad baron, all of whom, by lucky happenstance, were "light" on the land, allowing much of its story to remain undisturbed.

And to learn what the inhabitants of ancient Sardis may well have considered to be their form of homeowners' insurance, read "Artifact" (page 68).

Claudia Valentino

Claudia Valentino
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The Grand Challenges

Archaeologists spend a surprising amount of time in the “present”—conducting a survey along the track of a new gas pipeline, or excavating a site threatened with destruction, for example. Rarely are they able to take a step back and think about the entire human career as revealed by archaeology. Now 15 leading American archaeologists have done just that. With support from the National Science Foundation, they have asked the archaeological community to help them define the “grand challenges” in archaeology today. In a paper published in the January 2014 issue of the journal *American Antiquity*, they have posed a series of questions that will attract significant research attention in the coming decades.



What are some of these questions? Underlying all of them is the continuing interplay between humans and the environment. The environment has conditioned human behavior, and humans, through their activities, have shaped the world in which we live. Given this background, the first and most fundamental question is, how did humanity arise? Then, how have humans responded to changes in climate and the environment in the short term and the

longer term? How have they behaved when faced with abrupt, destructive natural events, such as earthquakes, tsunamis, volcanic eruptions, floods? All of these have had a strong impact on human societies at various times in the past.

Other questions concern population dynamics: What are the factors that drive or limit population growth? What has determined good health, disease resistance, and physical well-being among humans in the past?

These questions concern the long trajectory of human evolution from the beginning. Other questions focus on the last several thousand years. What led to the peopling of the continents by modern humans, and what were the consequences of this extraordinary migration story? Why did people develop agriculture? What social and demographic factors underlay the development of cities?

Conflict is a seemingly ever-present element among societies today. What role did conflict play in the development of complex societies in the past? We know, too, that most human societies, whether simple village communities or extensive civilizations, have, in time, collapsed or at least undergone major transformations. Are there common factors that have contributed to these events? Can we identify patterns in them, and what might this tell us about the sustainability of human societies in the future?

These are large questions indeed, and of increasing importance for understanding communities in the present as well as in the past. We look forward to learning more about how human societies developed—and about ourselves—as archaeologists tackle these issues in the years to come.

Andrew Moore

President, Archaeological Institute of America

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How to Outsmart a Millionaire

Only the "Robin Hood of Watchmakers" can steal the spotlight from a luxury legend for under \$200!

I wasn't looking for trouble. I sat in a café, sipping my espresso and enjoying the quiet. Then it got noisy. Mr. Bigshot rolled up in a roaring high-performance Italian sports car, dropping attitude like his \$14,000 watch made it okay for him to be rude. That's when I decided to roll up my sleeves and teach him a lesson.

"Nice watch," I said, pointing to his and holding up mine. He nodded like we belonged to the same club. We did, but he literally paid 100 times more for his membership. Bigshot bragged about his five-figure purchase, a luxury heavyweight from the titan of high-priced timepieces. I told him that mine was the **Stauer Corso, a 27-jewel automatic classic now available for only \$179.** And just like that, the man was at a loss for words.

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LETTERS

Update From 716 Fathoms

There have been some exciting revelations to report on the Monterrey wrecks ("All Hands on Deck," March/April 2014). For those worried about the fate of the ceramic bowl that *Herc's* manipulator appeared to pummel, they'll be pleased to know it survived, unbroken. After performing well, *Herc* suffered a hiccup when the ambient sea pressure formed a bubble in one of the oil-filled hydraulic lines running the arm. Fortunately, each punch hit soft mud and missed the bowl and other artifacts.

Meanwhile artifact analysis and conservation continues. The T-shirt, the only modern artifact seen on the wreck so far (and there is always mod-

ern trash on wreck sites), is the first item to emerge from conservation—a simple soft-cycle wash. It turns out to be a coarse fabric tee with the logo of a Manila (Philippines) construction company. And in other exciting news, we discovered two new hitherto undocumented life forms, another reminder that the deep ocean truly is Earth's final frontier, capable of yielding information about new life as well as past civilizations. To read more, go to www.archaeology.org/monterrey

James Delgado

*Director of Maritime Heritage, National Oceanic and Atmospheric Administration
Washington, D.C.*

A True Gentleman

I just read "All Hands on Deck" (March/April 2014). The Monterrey shipwrecks are an intriguing story and Lauren Hilger's coverage was very well done. I must, however, respectfully disagree with her characterization of Jean Laffite. Laffite was many things, including a legal

privateer and sometime pirate. He was also a U.S. war hero at the Battle of New Orleans, where he possibly turned the tide for the Americans. He was a smuggler, a tax evader, a spy, and a very savvy businessman. One thing he wasn't was "ruthless." He was never accused of violence, and was known among his contemporaries, including the authorities, as a gentleman.

Jack Watson

Galveston, TX

Roman "Civilization"?

"Ham Hill's Violent History" in the January/February 2014 issue, an article on the massacre of "hundreds, if not thousands" at Ham Hill, Britain, attributes the massacre to the Romans. Later in the same issue, in "An Epic Conflict," we learn of the slaughter of the inhabitants of New Carthage, Spain, also by the Romans. How common was massacre a standard Roman military tactic? Has anyone compiled a history of this Roman terror?

Dan Dorritie

Sacramento, CA

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From the Trenches

LATE-BREAKING NEWS AND NOTES FROM THE WORLD OF ARCHAEOLOGY

A Brief Glimpse into Early Rome

The Sant'Omobono archaeological site in Rome isn't much to behold. It is strewn with a seemingly haphazard array of stone blocks, walls, ditches, and the occasional column drum—hardly pleasing to the eye. Even the eponymous Renaissance church that sits atop the site has been labeled as “incongruously ugly” by one prominent archaeologist. So it's not surprising that tourists on their way to better-known monuments scarcely give it a second look.

This past January, however, the site suddenly received a great deal of attention when multiple international news sources reported incorrectly that Rome's oldest temple had just been discovered there. As the archaeologists of the Sant'Omobono Project, an ongoing collaboration between the Superintendency of the Capitoline, the University of Calabria, and the University of Michigan, had actually made clear, this temple has been known—although rarely seen—since the 1930s, when it was first discovered.

Beneath Sant'Omobono lies a series of religious buildings that defined the site for much of the first millennium B.C. The importance of this area, situated at a bend in the Tiber River and known as the Forum Boarium, or “cattle market,” stemmed

from its location at the nexus of several ancient communication and trade routes. It was the economic center of early Rome, complete with the city's earliest river harbor capable of providing a safe haven for ships, and an international marketplace where goods, as well as cultural ideas, religions, and languages, were exchanged. It was also a prime location for the construction of a temple to protect and oversee this activity.

“The [Sant'Omobono] site is crucial for understanding the related processes of monumentalization, urbanization, and state formation in Rome in the late Archaic period,” says Dan Diffendale, a member of the University of Michigan team. The site's main archaeological features are the twin temples of Fortuna and Mater Matuta, which date to the late sixth or early fifth century B.C. Although these are among the oldest temple remains in ancient Rome, they actually sit atop an even older structure. This is the ancient Archaic period (roughly 800–500 B.C.) temple that January's reports referred to.

This structure has been notoriously difficult to evaluate, because it lies deep beneath the ground, well below the city's water table. Now, however, the Sant'Omobono Project has finally been able to get a glimpse at the early temple's foun-

Sant'Omobono archaeological site, Rome



Retaining walls, Sant'Omobono



Oldest known temple remains in Rome (7th–6th century B.C.)



dition. In order to achieve this, steel retaining walls were hammered into the ground to hold back the weight of the waterlogged soil, and mechanical pumps were then used to drain the

trench. At a depth of more than fifteen feet, three courses of stone blocks belonging to the ancient temple were identified. “We were able for the first time to record the Archaic foundation with modern means, using total station and photogrammetry, as well as extensive photography,” says Diffendale. “This is the first time that the Archaic podium has been exposed in relatively dry condition.”

In addition, the Sant’Omobono team was able to recover hundreds of artifacts, including votive offerings, drinking vessels, and figurines. These

suggest that the temple dates to the late seventh and early sixth centuries B.C., making these the oldest known temple remains in Rome.

For safety reasons, the trench could stay open for only three days before it was backfilled. Nevertheless, this latest round of work will have long-lasting effects on archaeologists’ study of the development and growth of the eternal city. The Sant’Omobono site has finally gained its rightful place in the spotlight as one of ancient Rome’s most important, even if least well known, sites.

—JASON M. URBANUS

OFF THE GRID

More than 1,200 years ago, a group of Caddo Indians called the Hasinai established a village in present-day Cherokee County, Texas, where they found good soil, a teeming forest, and nearby springs. Today the site, known as Caddo Mounds State Historic Site, is one of the best known and intensively investigated Native American sites in Texas. It is the southwesternmost ceremonial center of the Mound Builder culture, who flourished in the woodlands to the east from 1000 B.C. to A.D. 1550. The site was abandoned in the thirteenth century, but its spirit lives on in the modern Caddo Nation based in Oklahoma and in the name “Texas,” which comes from a Caddo word meaning “friend.” Archaeologist Brett Cruse of the Texas Historical Commission says, “It was

a major early Caddo center, and perhaps no other Caddo site has had as great an impact on what we know about them.”

The site

Visitors today can see three earthen mounds and a large portion of where the village was situated, estimated at more than 110 acres. Two mounds are platform temples that were used for political and religious ceremonies. The third is a burial mound containing several dozen tombs of community leaders, some of whom appear to have been accompanied by sacrifices. The associated village includes residences of common people, granaries, and garden plots. Archaeologists have learned a great deal about Caddo culture

from excavations at the site. For example, residences in the village, dating to as early as A.D. 850, were organized into compounds with small courtyards. The form persisted for several hundred years, demonstrating great continuity in Caddo community and social organization. Within special areas near the temple mounds, excavators have also uncovered evidence of a series of large public buildings that probably served as temples, council or clan houses, and perhaps residences of community leaders. Outside these areas were smaller houses that had been built and rebuilt in the same place for generations. A self-guided tour of the site can take about an hour, and there is a visitor’s center museum that exhibits about 200 artifacts, including pottery, tools, and weapons.



Caddoan effigy pipe

While you’re there

Six miles west of Caddo Mounds is Mission Tejas State Park, a 660-acre park with camping, hiking, and fishing, as well as a representation of Mission San Francisco de los Tejas, the first Spanish mission in the province of Texas, established in 1690. Just 18 miles away is a depot of the Texas State Railroad, which runs historic train engines and cars several days a week. The historic downtown of Nacogdoches, 26 miles away, offers several historical museums, food, shopping, and nightlife.

—MALIN GRUNBERG BANYASZ



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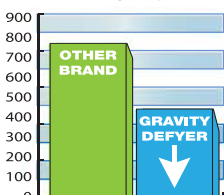
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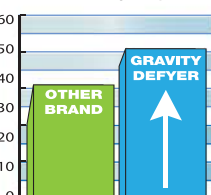
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Port of the Pyramids

Sometimes great discoveries are made when archaeologists stop digging and take a closer look at what has already been found. This is what happened to Mark Lehner, director of Ancient Egypt Research Associates, as he was looking over maps of Heit el-Ghurab, where he has worked for 25 years. Lehner had thought that Heit el-Ghurab, a 4,500-year-old town next to the pyramids of Giza, was home to the workers who built the pyramids. But now he believes the town was also a wealthy Nile River port. While examining the distribution of wood and pottery in a part of the town where laborers lived, Lehner recognized that much of it came from the Levant. He surmised that the laborers who built the pyramids were probably also sailors and that Heit el-Ghurab was probably the southern end of an important trade route called the Byblos Run, which brought lumber, olive oil, and other trade goods from territories as



far north as modern-day Lebanon. “It’s a whole consilience of different lines of evidence that brought us to this aha moment,” says Lehner. This rich port also had its own elite class. In the western part of the town, Lehner’s team found leopard teeth, which were worn as ornaments by certain Egyptian priests. The archaeologists also uncovered the bones of calves—veal would have been an especially prized luxury.

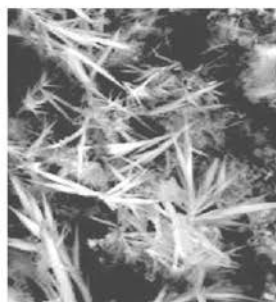
—ZACH ZORICH

Hardening Brittle Bones

Just like the bones of living people with osteoporosis, human remains in the archaeological record can lose bone mass over time. These samples usually require treatment with plastic fillers before they can be moved and studied.

A team of chemists at the University of Florence has devised a novel alternative that uses a chemical reaction to return lost minerals, such as calcium, to fragile old bones. The scientists treated fragments of degraded bone dat-

ing from A.D. 1000–1400 with a solution of calcium hydroxide nanoparticles in alcohol. Over several days, the nanoparticles reacted with airborne carbon dioxide and traces of collagen in the bone to form needle-like crystals of aragonite, a hard variant of calcium carbonate that forms naturally in seashells and corals. The crystals left the bones—in this case a saint’s relics from



the Basilica of San Clemente in Rome—smoother, whiter, and much stronger.

The method, says study coauthor Luigi Dei, requires further investigation. Unlike the use of plastic fillers to shore up bone, this process is likely irreversible. Once perfected, however, it could be of use wherever archaeologists encounter brittle bones.

—KATHERINE SHARPE

First American Family Tree

The burials of two boys—each found decades ago and thousands of miles apart, but recently subjected to genetic analysis—are helping settle the matter of where the first residents of the Americas came from.



Clovis tools from the grave of Anzick boy

A lab at the University of Copenhagen, led by geneticist Eske Willerslev, has sequenced the genome of a four-year-old boy whose remains were found near the Siberian village of Mal'ta in the 1920s. Dating back 24,000 years, the boy's DNA was found to be closely related to that of modern Native Americans, suggesting that some of the Mal'ta people's descendants migrated to the New World. Researchers were surprised that he was significantly related to Europeans, as well as South and Central Asians, but not to East Asians, who have long been known to be connected to Native Americans. The scientists believe that today's Native Americans are a mixture of the Mal'ta and East Asians they encountered en route from Siberia to the Americas. "The genetics tell us about homelands," says Mike Waters, a Texas A&M University archaeologist involved in the research, "but it doesn't tell us about routes."

Across the Bering Strait, the remains of a one- to two-year-old boy, called Anzick boy, discovered in western Montana more than 45 years ago, are the oldest known burial in the United States, dating to 12,500 years ago. The same Danish lab studied his genome and found that he is a direct ancestor of all modern Native American populations from Mexico to

Chile. The boy is also closely, but less directly, related to tribes in Canada and the Arctic. However, because of distrust between the U.S. Native American community and scientists, not enough DNA of American tribes is available to determine their relation to the boy. Waters predicts that "a large chunk of Native Americans in the U.S. will show a strong

direct ancestry to the Anzick boy."

Willerslev says that sequencing more ancient genomes from around the Americas might eventually uncover more detailed migration patterns. "We all have years of work ahead of us if we would like to understand the full picture of the early peopling of the Americas," he says.

—NIKHIL SWAMINATHAN

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Recreating Nordic Grog



Museum display of a woman's grave from Juellinge, Denmark, with Roman wine set and strainer cup

The woman, dead at 30, was buried 1,900 years ago in an oak log near Juellinge, Denmark. Interred with her was a long-handled bronze strainer that still held residue of a fermented drink she may have been meant to enjoy in the afterlife.

Now the ingredients and even the flavor of that drink, a “grog” made from local fruits, grains, and herbs mixed with grape wine from southern Europe, are becoming clearer. University of Pennsylvania archaeologist Patrick McGovern has applied biomolecular techniques to organic residue taken from four ancient Scandinavian artifacts, including the woman’s strainer, a clay jar, and pieces of Roman bronze

berry, bog cranberry, rye, barley, juniper, birch, pine, bog myrtle, and yarrow. Tandem mass spectrometry then showed the presence of tartaric acid, the biomarker for wine.

“This work is the first to prove that wine was being traded from the south to the north at this time,” says McGovern. It has also created a detailed, consistent picture of ancient Scandinavia’s preferred beverage of distinction. McGovern is working with Delaware’s Dogfish Head Brewery—as he has on previous concoctions based on ancient residues—to create a modern rendition of the sour, fruity, herbaceous grog.

—KATHERINE SHARPE

drinking sets, dating to between 1500 B.C. and A.D. 200.

Using a method called solid phase micro-extraction, McGovern found volatile organic compounds that are biomarkers for ingredients such as lingon-

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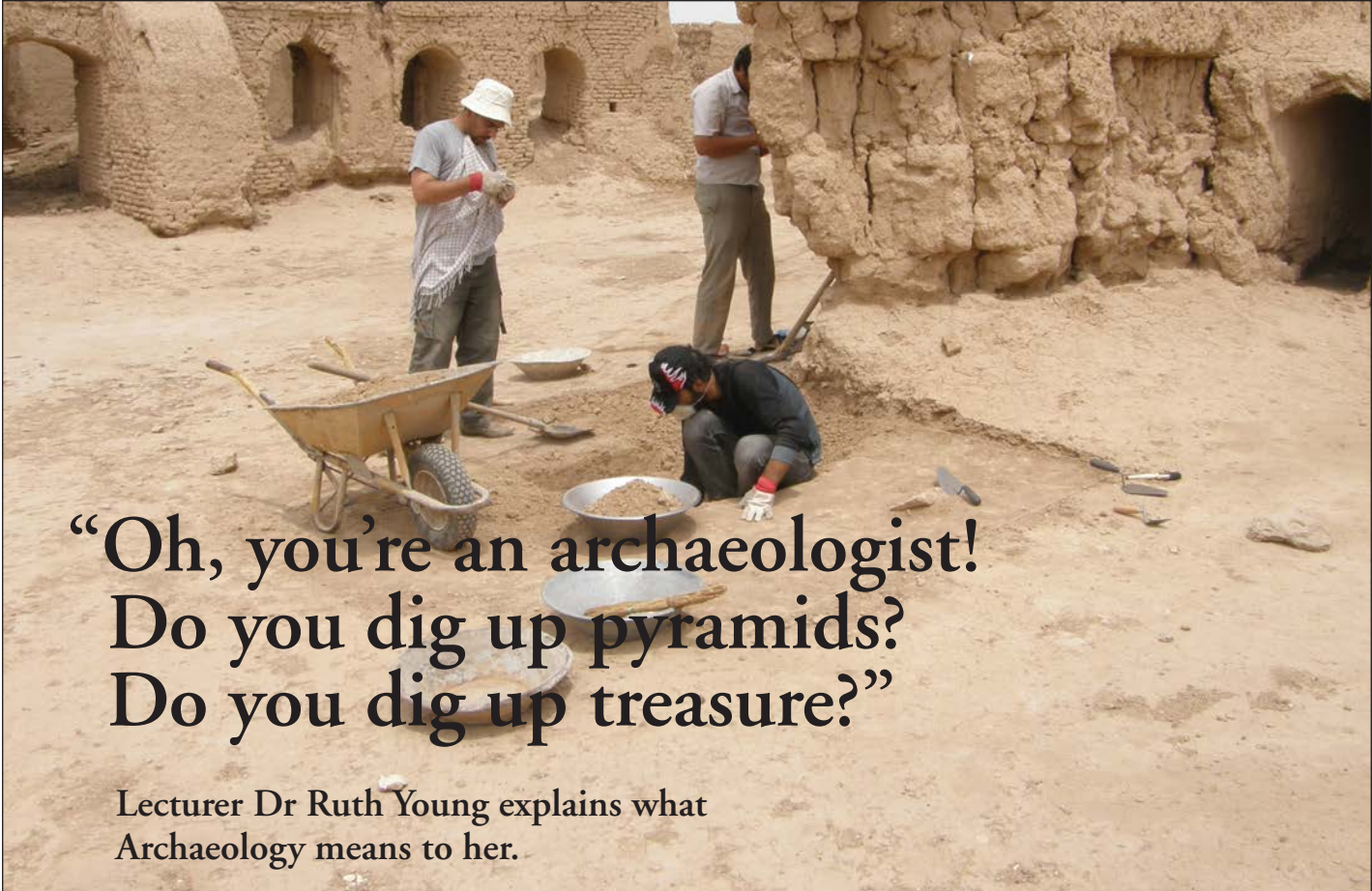
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“Oh, you’re an archaeologist! Do you dig up pyramids? Do you dig up treasure?”

Lecturer Dr Ruth Young explains what
Archaeology means to her.

This is a familiar response whenever I tell people what I do for a living (usually on a plane), and I think that they are often a little bit disappointed when I tell them about the reality of my work: currently recording mud brick villages in Oman from the recent past, and linking the structures to the memories of people who used to live in them. While people can immediately grasp the scale of the pyramids, and totally be behind digging up gold and rubies, figuring out why anyone is interested in old rubbish and ordinary people is a bit harder.

Yet to me, the great joy of archaeology is learning about ordinary people of the past and finding out more about how they lived, why they made the decisions they did, how they adapted to big social changes, and what their place in the world might have looked like. Archaeologists do this through studying material culture – which usually is the old rubbish of the past, whether it’s abandoned buildings from the 20th century, or a site like Persepolis. At the School of Archaeology and Ancient History at the University of Leicester, we bring our passion for our subject and our world-class research to all our teaching; not only our campus-based programmes, but also our world-leading Distance Learning programmes that mean you can study towards a BA, MA or PhD from anywhere in the world. Discover more at www.le.ac.uk/archaeology

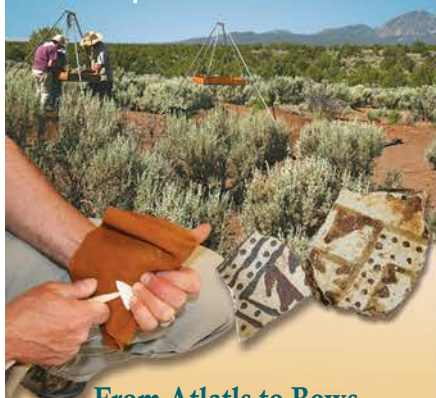
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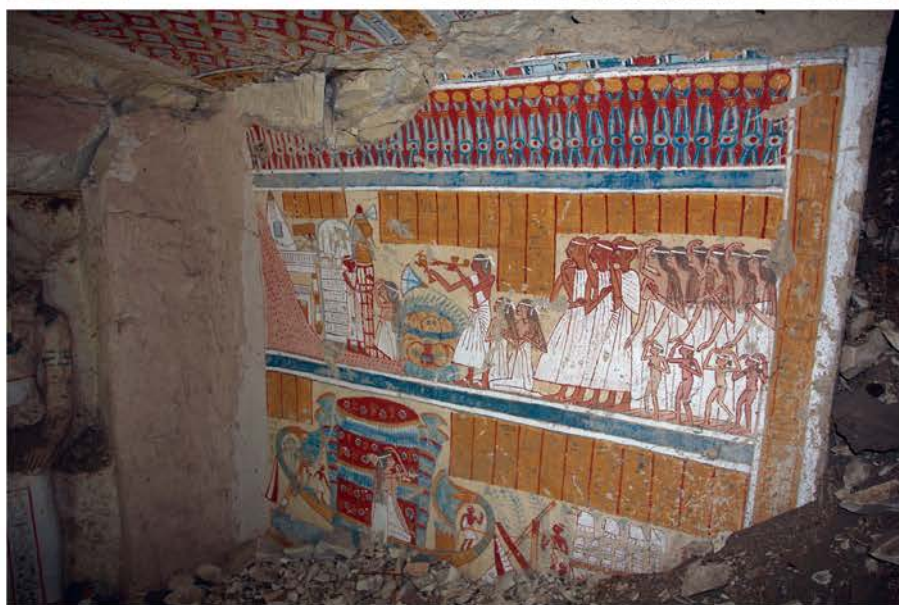


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FROM THE TRENCHES

The Goddess' Brewer



A team of Japanese Egyptologists doing routine cleaning at a New Kingdom necropolis in Luxor accidentally uncovered a previously unknown burial chamber decorated with still-vivid murals. "I was surprised to find such a beautiful tomb," says Waseda University's Jiro Kondo, who points out that only a handful of chambers with such well-preserved murals have been unearthed at the necropolis. Among the paintings, which date to between

1292 and 1069 B.C., is a depiction of the funeral procession of the tomb's owner, an official named Khonsuemheb. According to inscriptions in the chamber, he was the chief beer maker of the temple of the mother goddess Mut—an important ritual position. On the ceiling, two figures of Khonsuemheb are painted next to the text of a hymn to the sun god Amun-Ra and a depiction of a solar boat, in which the dead were thought to sail for eternity.

—ERIC A. POWELL



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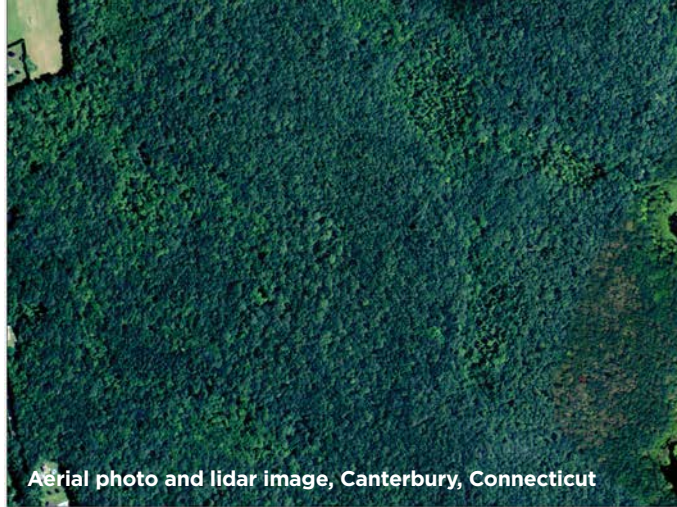
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Peeping through the Leaves



Aerial photo and lidar image, Canterbury, Connecticut

Southern New England's forests are beloved for the bright colors of their fall foliage, but they also hide much of the rich record of human influence on the historic landscape. Using lidar, which employs lasers to see beneath tree cover, Kate Johnson of the University of Connecticut has been able to peer into the past of the Northeast.

"We have identified old stone walls, foundations, dams, mills, and abandoned roads and paths that are obscured in aerial and satellite imagery," says Johnson. "Being able to see these features on the landscape is exciting because this type of data was never available before."

—JARRETT A. LOBELL

Clash of the War Elephants

Elephants were the tanks of ancient Mediterranean warfare, commonly used for trampling and intimidating enemies. Yet there was only one battle in which African elephants and their Asian cousins are known to have met—the Battle of Raphia, in Gaza, in 217 B.C. According to the historian Polybius, it wasn't even a contest. He writes that the African pachyderms, under the command of the Ptolemaic pharaoh of Egypt, panicked and tried to flee at the sight of the larger Asian elephants of the Seleucid army. Yet African savannah elephants are typically bigger and stronger than Asian ones. Had Polybius gotten it wrong? Modern writers have speculated that the Egyptians had African forest elephants, a smaller species than the savannah variety. Later Roman and Carthaginian armies, including Hannibal's forces, might also have used forest elephants.

That idea has crept into modern accounts, depictions, and even video games such as *Age of Empires*, in which war elephants have the rounded ears and dwarfish proportions of the forest dwellers. Now geneticists have found that the species of elephant that the Egyptians had access to came from modern-day Eritrea in East Africa, and share no genetic markers with



African elephant



Asian elephant

forest elephants. "The idea that they used forest elephants was not based on evidence, but it got repeated over and over," says geneticist Alfred Roca of the University of Illinois at Urbana-Champaign. "It makes no sense. Forest elephants lived in the Congo basin, thousands of miles from the Mediterranean." The matter of Polybius' account remains unsettled.

—ROGER ATWOOD

England's Oldest Footprints

Nearly a million years ago, a small group of hominins walked near an estuary on the English coast 140 miles northeast of London, leaving footprints in the soft mud, which then hardened. Last summer, archaeologists working at the nearby site of Happisburgh, which dates to around the same time, discovered those footprints. They are the oldest footprints known outside of Africa—49 of them, made by at least five individuals, including some children. “One can imagine a small family group walking along the edge of the estuary looking for seaweed or shellfish,” says Nicholas Ashton of London’s Natural History Museum. Some of the footprints were lifted out of the site in a block of sediment before they could be eroded away by the tide. The sediment will be CT scanned to provide a more detailed picture of the stature and gait of the people, probably *Homo antecessor*, who made the prints.

—ZACH ZORICH





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Secrets of Bronze Age Cheese Makers



Bronze Age mummies in China's Taklamakan Desert were buried with the earliest known pieces of cheese affixed to their necks and chests. Chemist Andrej Shevchenko of the Max Planck Institute of Molecu-

lar Cell Biology and Genetics in Dresden came to that conclusion after analyzing several enigmatic pieces of yellow matter found on the mummies,

which date to between 1980 and 1450 B.C.

Shevchenko was able to isolate proteins characteristic of cheese in the samples, and was even able to identify the kinds of yeast and bacteria used in the fermentation process. "That's what makes this so exciting," says Shevchenko. "We know that cheese was made as early as the sixth millennium B.C., but this is the first evidence that shows just how people were making it." Shevchenko points out that just 20 years ago organic matter such as these pieces of cheese might have been ignored and disposed of. "With the technology we have now, no one should throw out anything," says Shevchenko.

—ERIC A. POWELL

Our Lady of the Lake

Archaeologists were surprised by the discovery of a Byzantine basilica in just six feet of water, 60 feet from the shore of Turkey's Lake Iznik, near the ancient city of Nicaea. Locals knew there were ruins in the lake, but scholars were unaware of the 100-by-60-foot structure until Uludag University's Mustafa Sahin recently spotted the clear footprint of the church in an aerial photo taken by the local municipality. Sahin thinks it's likely the church was built in the late fourth or early fifth century, less than 100 years after Nicaea hosted the First Ecumenical Council in A.D. 325, convened by the emperor Constantine to establish key doctrines for the Eastern Roman Empire.

—JENNIFER PINKOWSKI



Big Data, Big Cities

Ancient cities such as Teotihuacan may have had more in common with today's sprawling metropolises than scholars previously imagined. University of Colorado Boulder archaeologist Scott Ortman and his colleagues have found that equations used to describe growth patterns in today's urban centers also work when applied to early cities.



Teotihuacan, Mexico

Scientists developed these mathematical models to predict how population growth affects a modern city's population density, infrastructure, and economic production. But Ortman noticed that while the models depend on a number of variables, such as the rate of social interaction and the size of the settled area, none relied on a specific modern technology. "I thought that if this approach had any legs at all then it ought to apply to any human society, past and present," he says.

If he is correct, then ancient settlements would have become denser in a mathematically predictable way as their populations grew. To test the theory, Ortman and his colleagues analyzed archaeological data collected during surveys of some 1,500 settlements throughout the Valley of Mexico. These ranged from small farming hamlets dating to 1500 B.C. to the Aztec capital of Tenochtitlan, which, at its height around A.D. 1500, was home to some 200,000 people. The team found that, just as with modern cities, the equations accurately predicted the relationship between set-

tlement size and population density.

According to Ortman, this insight means that archaeological data could have a role to play in modern policy debates surrounding issues such as the economics of rapid urbanization. "There's a tendency for social scientists to think that the Industrial Revolution made all prior human history irrelevant

to what's going on now," says Ortman. "But if the basic processes that underlie urban growth phenomena have been the same throughout time, then the archaeological record is a store of human experience that has something to offer public discourse about the issues we face as we move into the future."

—ERIC A. POWELL

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CALIFORNIA: Bitumen, or natural petroleum tar, has been a useful material around the world for thousands of years. In the Channel Islands, the Chumash used it as long as 8,000 years ago as an adhesive or to waterproof baskets or boats, but the islands have no terrestrial source. They could have traded for some, but scientists applied geochemical analysis and traced samples used by the Chumash back to a petroleum seep 25 miles offshore. The bitumen washed ashore in the form of “tar balls” or “tar whips.”



NEW YORK: In 2010, in a heap under City Hall Park in lower Manhattan, archaeologists found alcohol bottles, pipes, and bones, including those of turtles, which would have been a delicacy. The soiree deposit dates to the early 19th century, right around when City Hall was built. Researchers have only just determined the identity of an odd cylindrical device carved from bone, with a perforated cap. Comparisons with museum objects reveal it to be a syringe for feminine hygiene—a douche.

DENMARK: Fresh oysters are worth the elbow grease and tricky leverage needed to pry them open, but that's not the only option. The thick Mesolithic middens near the village of Ertebølle are full of oyster shells, but bear no evidence of tools to open them or damage from the process. Rather, some show signs of light scorching. Experiments show that after brief heating on hot stone or embers, oysters ease open with a puff of steam. But the researchers involved disagree on whether the process affects the taste.



GUADELOUPE: Coastal erosion on the island of Grand-Terre has put a colonial-era graveyard at risk. The beachside cemetery has been known since the

1990s, when a skull with a slave collar surfaced. Archaeologists have now revealed almost 50 graves, including an individual with cut incisors, suggesting African origins. It is estimated that hundreds more graves remain, as the cemetery was used for around a century. The French territory had a large population of slaves working on sugar plantations until the mid-19th century.

PERU: A new study of dozens of skulls from the south-central Andes examined the practice of trepanation, or the surgical removal of a portion of the skull. Researchers theorize that the practice arose following the collapse of the Wari Empire around A.D. 1000, which may have brought increased violence and health problems that required new, and sometimes radical, treatments. The study shows that primarily adult men were eligible, that scraping offered better outcomes than drilling, and that some skulls were used for practice after death.



MOROCCO: Tooth decay occurs when bacteria consume trapped food and produce acid that breaks down hard dental tissue. These bacteria thrived when humans took up farming;

ancient hunter-gatherers appear to have had few cavities. Not so the people of Grotte Des Pigeons 15,000 to 12,000 years ago. New analysis of dentition shows that they had as many cavities as modern populations do. This could have been due to a diet heavy in acorns and other carbs that are appealing to the agents of tooth decay.



FINLAND: Metal detectorists found a fascinating grave in a field in Janakkala and turned the find over to archaeological authorities. The well-preserved body dates to the Northern Crusades, in the 12th and 13th centuries, when

Christian Swedes and Danes invaded pagan areas of Northern Europe. The individual, perhaps a well-off soldier, was buried with two swords: a broken Viking-era piece and an intact one from his own time. At nearly four feet from pommel to point, it one of the longest ever found in Finland.

CHINA: Experts have revealed a multiplication table that might have been used to calculate land area, crop yields, and taxes during the Warring States Period, around 305 B.C. The table was among 2,500 bamboo strips that contain many significant historical texts donated to a university in



Beijing. Twenty-one of them contain numbers and, when properly arranged, allow for the multiplication of whole and half integers from 0.5 to 99.5. It is among the oldest and most sophisticated tabular calculators known.



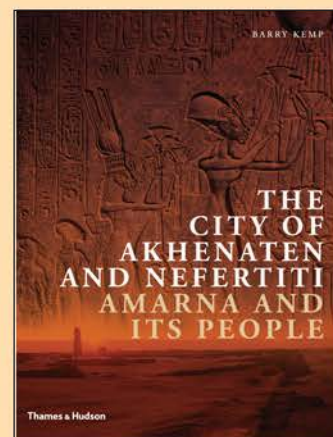
IRAQ: The Royal Diary of Alexander the Great describes

his death in Babylon in 323 B.C. as an agonizing 12 days of fever, weakness, pain, and vomiting. The latest theory on the cause of his demise is poisoning with a plant called white hellebore. Toxicologists found that hellebore poisoning matches many of Alexander's symptoms. Others disagree, stating that the symptoms would have been recognized by his doctor—unless, of course, his doctor had been part of a murder conspiracy (as legend has it).



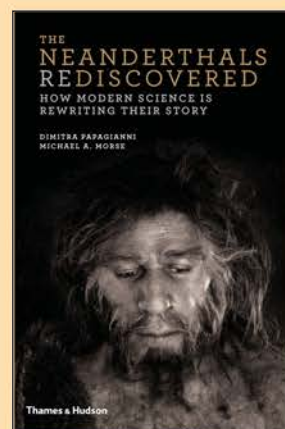
INDIA: Queen Ketevan ruled a small Christian kingdom in eastern Georgia. After her death in Iran in 1624—she was tortured after refusing to convert to Islam—her remains were unearthed and her right arm was taken to a convent in Goa. The convent had fallen into ruins by the 19th century, but excavations turned up bone fragments. Scientists found a genetic signature in the bones that is found in Georgia but not India, suggesting they have identified some portion of the lost martyr's relic.

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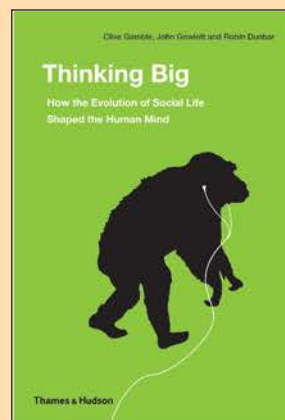
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The pharaoh Akhenaten built his new capital of Amarna (above) more than three millennia ago in an uninhabited stretch of Egyptian desert. In less than two decades, a city filled with palaces and temples was completed. Recently, archaeologists have excavated in an area known as Grid 12 and found the modest homes (far right) of the workers who built the city.



BEYOND THE PALACE WALLS

**Nothing is quite what it seems in
Egypt's ancient city of Amarna**

by **HEATHER PRINGLE**



TODAY, THE ANCIENT EGYPTIAN city of Amarna is in ruins, its rubble scattered over a desert plain. The stately halls of its temples are gone, its crumbled residences buried beneath windblown sand. The elite tombs carved in cliffs to the east and painted with scenes of elegant court life remain half finished, as they were in antiquity. The view was very different some 3,350 years ago after the iconoclastic 18th Dynasty pharaoh Akhenaten chose this strip of wilderness along the Nile as the site of his new city.

Motivated by the need for a pristine landscape unsullied by the worship of old gods, Akhenaten moved the capital of

Egypt some 170 miles northwest from Thebes, where it had been for much of seven centuries, to a sacred territory he called the “Horizon of the Sun’s Disc.” There he built palaces for himself and his family, and temples and shrines dedicated to the pure power of the sun, a force known as the Aten, the creator of all life. Turning his back on Egypt’s busy multitude of gods—and on more than a millennium of tradition—the pharaoh attempted to cleanse Egypt’s spiritual traditions of what he is thought to have viewed as distortions and human inventions. And to build his new capital city, Akhenaten uprooted perhaps as many as 40,000 people from their homes elsewhere in Egypt and brought them to a harsh land and the lives of toil

that were required to build and maintain this 1.5-square-mile city.

Yet Akhenaten's reign endured for only 17 years. His successors, including his son Tutankhamun, shunned his revolution and returned to the old capital at Thebes. The pharaoh's name was purged after his death from official records, and both his singular religious vision and impressive city were abandoned, then largely forgotten. It was not until the mid-nineteenth century that archaeologists recognized Amarna as a royal capital, and then identified its builder as Akhenaten, a pharaoh then lost to history. In Amarna, archaeologists found fragmentary murals and ancient reliefs that portrayed it as a place of earthly delights, with lush royal gardens, riverbanks brimming with birds and butterflies, pro-



Breaking with tradition, Akhenaten moved Egypt's capital from Thebes to Amarna.

cessions of sleek-looking citizens, and temple tables heaped with bread, wine, and game. But was Amarna really as advertised by its artists—a city of beauty, wealth, and plenty?

In 1977, a young University of Cambridge Egyptologist, Barry Kemp, arrived in Amarna to begin a detailed mapping project of the city's many weathered features. Returning nearly every year after that, Kemp launched an investigation of the city's ancient topography, urban environment, houses, and cemeteries. Today, more than three decades later, Kemp and a team of archaeologists, physical anthropologists, bioarchaeologists, botanists, zoologists, and entomologists are



In this relief, artists have depicted Akhenaten, his wife, Nefertiti, and their daughters with elongated heads and curving bodies in a style that represents a radical break from traditional Egyptian art. At the center of the relief is the sun, the singular deity upon whom the pharaoh focused his personal religion.



Many paintings from the city, such as this one from Akhenaten's palace, portray Amarna as a place of abundance brimming with flowers, plants, and birds.

revealing a new view of Amarna. Far from living in prosperity, the city's poorer inhabitants endured hunger and malnutrition, performed backbreaking manual labor, and bore the signs of a generally grim life, says Kemp. Moreover, despite its grand temples and palaces, Amarna was hardly the glossy showplace portrayed in the ancient murals. Its residential areas were packed with workshops and production centers, forming "nothing less than a giant factory," according to Kemp.

KEMP, NOW A PROFESSOR emeritus at Cambridge, was first drawn to Egyptology as a child when he saw a collection of photographs his father had taken of Luxor and other sites while he served in the British army during World War II. Later, as a young university student, Kemp became fascinated by ancient Egyptian cities and towns and how they had functioned. Amarna quickly caught his eye. Ancient Egyptian texts suggested that the capital was occupied for less than two decades, during which very little was torn down or built over, leaving the original urban plan unobscured. It seemed an ideal place to study the components and workings of an early Egyptian city.

During his first field seasons at Amarna, Kemp puzzled over what he regarded as the city's unusual features. In the 1950s, urban sociologist Gideon Sjoberg, now a professor emeritus at the University of Texas at Austin, had published research showing that preindustrial cities tended to follow a particular plan. By studying the accounts of travelers who had visited cities as diverse as Cairo, Seoul, and Mecca during the Renaissance and later, Sjoberg found that builders tended to construct the seats of power—palaces, temples, and elite homes—in the central core. Artisans and craftspeople, often divided into guilds, took up residence in neighborhoods wrapped around the core. Finally, along the fringes, either just inside the city wall or banished to outlying suburbs, the poor congregated in

meager dwellings. The farther one lived from the seat of power, Sjoberg concluded, the lower one stood in the city's pecking order.

Yet Amarna, an early preindustrial city, wasn't like this. There was no city wall and there were no neighborhood guilds. And although the pharaoh's Great Palace and temples rose side by side in the center block, the city's elites built their villas elsewhere. Indeed, as Kemp pored over the maps and descriptions of earlier excavations, he found large villas scattered all over the capital, even along the outskirts, many of which had the names and titles of their owners—Akhenaten's high officials—inscribed on the doorframes. Kemp also noticed that adjoining each of the great villas was an intricate warren of small houses and courtyards. He wondered who lived in those houses and how they might relate to the wealthy officials.

Early archaeologists had dug large swaths of the capital rapidly, leaving relatively few intact residential areas in the area known as the Main City, which lies between the Nile River and the cliffs to the east. Kemp, while reopening an early-twentieth-century excavation in the Main City to obtain samples for the entomologists and other researchers, spied an expanse of rolling ground adjacent to the large home of Akhenaten's chief charioteer, a man named Ranefer. "The surface was relatively smooth and undulating, indicative of the presence of small houses," recalls Kemp. The area, designated Grid 12, appeared to contain an undisturbed community—exactly the kind of place that could reveal what was going on in the courtyards. "If we could ask Akhenaten which archaeologist he would choose to write his legend, he would say Kemp," jokes Zahi Hawass, Egypt's former Minister of State for Antiquities. "He has eyes like X-rays and he walks for days before he excavates."

Discovered in a large building in Amarna known as the King's House, this fresco fragment depicts two of Akhenaten's young daughters perched on cushions in an opulent palace room.



Kemp and Anna Stevens, an archaeologist at the McDonald Institute for Archaeological Research in Cambridge, began excavating the small houses in 2004. Over two field seasons, the team slowly exposed dozens of short, choppy, irregular mudbrick walls, and they found that each dwelling seemed designed for an individual family. The layout of each was somewhat different from the others, and the rooms came in a variety of sizes. And each of these homes neatly adjoined another in a kind of do-it-yourself community that likely grew swiftly—perhaps in just a few days—with every member cooperating. Intrigued, Kemp turned to the floor plans that earlier excavators of Amarna had drawn of housing adjoining other big villas. Grid 12, he soon discovered, wasn't unique.

To Kemp it appeared that Akhenaten had taken little interest in planning Amarna's neighborhoods. Instead, much of the work was left to the communities themselves, and neighborhoods grew "by a process that is not dissimilar to larger squatter cities in the world today," he says. In settlements such as the *pueblos jóvenes* or "new towns" that now ring the outskirts of Lima, Peru, for example, poor migrants from the countryside occupy small plots of vacant land and quickly construct simple houses from mudbrick, pieces of cardboard, and tin sheets. "It is a very effective way to colonize the land," says Kemp. However, Amarna was different in two key respects. Unlike in Lima's *pueblos jóvenes*, Amarna's workers were not relegated to the city fringes. Rather, they lived cheek by jowl with Akhenaten's most powerful officials, from senior military officers and policemen to priests and royal sculptors all over the city.

In 1933, at the height of the Great Depression, two prominent Egyptologists published findings from their excavation of another cooperative, self-organized community in one of Amarna's suburbs, to the north of the Main City. Henri Frankfort and John Pendlebury characterized the intricate network of homes as "squalid" and dismissed the neighborhood as a "slum." When their team found parts of a faience necklace folded in linen and tucked away in one courtyard corner in the community, the two archaeologists immediately assumed it was stolen, the proceeds of an ancient crime. "That the new owner did not dare to display the compromising finery," wrote Frankfort and Pendlebury, "is suggested by the very circumstances in which we found it." Kemp believes this assessment stemmed from popular stereotypes of the day, which portrayed contemporary poor

urban neighborhoods as the dens of thieves and idlers. Grid 12 actually has a very different story to tell.

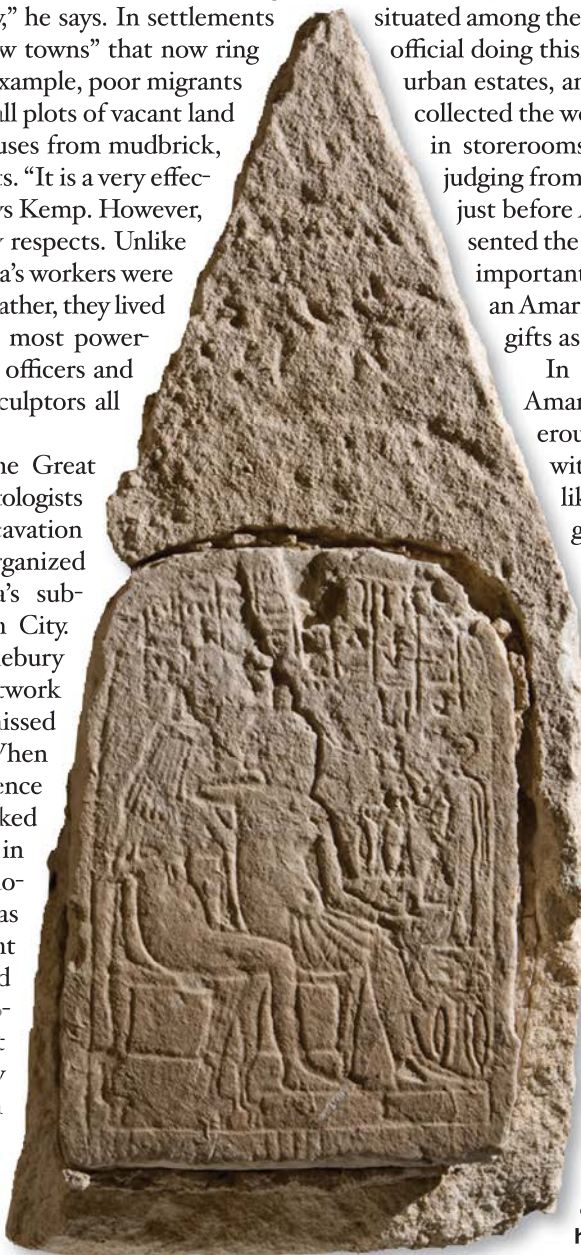
Inside the Grid 12 courtyards, Kemp and his team found debris from two industries. In some parts of the grid, craftspeople had worked at grinding and polishing small glass ingots that were used as decorative inlays. In another area, a scattering of small bits of turquoise-blue faience had been left behind—refuse from producing beads, pendants, and decorative tiles of the kind that adorned the royal palaces and temples. It seems that Amarna's squatters, rather than being thieves and idlers, were actually small-scale manufacturers.

The workers in Grid 12 manufactured the tiles and inlays for their patron, the senior official Ranefer. In all likelihood, it was Ranefer who had brought these workers to Amarna, and who provided them with land for their small dwellings. "They weren't slaves," explains Stevens. Rather, the workers were a vital part of Ranefer's estate, and his own home was situated among theirs. Ranefer wasn't the only high-ranking official doing this. Other dignitaries presided over similar urban estates, and they, or their overseers, seem to have collected the workers' finest goods and stockpiled them in storerooms or in small estate warehouses. Then, judging from scenes in several Theban tombs painted just before Akhenaten's reign, these dignitaries presented the warehoused goods to the pharaoh during important festivals. In a royal speech recorded on an Amarna tomb wall, Akhenaten describes these gifts as a form of levy or tax.

In several scenes carved on the walls of Amarna tombs, the pharaoh is shown generously rewarding his loyal senior officials with gifts. And in turn, his senior officials likely remunerated their workers with grain from their private fields. "You can start to understand how the situation worked," Stevens says. Far from being drains on the system—as Depression-era archaeologists once thought—Amarna's poorest communities were, in fact, the engines of production. "The whole city was there to serve the king and court," says Kemp.

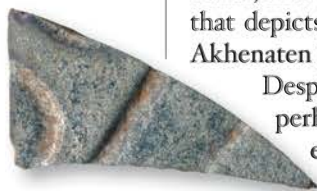
TO REFINE THIS NEW PICTURE of Amarna, Kemp needed to gather data from the inhabitants themselves, but more than a century of searching had failed to turn up any trace of commoners' cem-

In a workers' cemetery, archaeologists found this incised grave marker portraying a young couple seated side by side, a pose seen on reliefs depicting Akhenaten and his wife Nefertiti.





Recent excavations at Amarna have shown that the city's many poor communities were a vital part of a loosely structured factory system that turned out faience and glass inlays (left) to be used to decorate Akhenaten's temples and palaces.



eteries, even on the city's outskirts. In 1924, British Egyptologist Francis Llewellyn Griffith lamented, "There are square miles of sand waste leveled by wind and rain where a cemetery might lie without indications on the surface." Kemp, however, wasn't prepared to stop searching. More than a decade ago, his colleague Helen Fenwick, an archaeologist at the University of Hull, embarked on a new survey of Amarna's desert hinterland, recording and mapping all visible features. Over a three-year period, she discovered not just one cemetery, but five.

Kemp chose in 2006 to excavate in the so-called South Tombs cemetery, which lay along the sandy banks of a long-dry streambed. At some point, whether in antiquity or more recently, thieves had rifled through many of the burials there, likely searching for metal artifacts. In their haste or disappointment, they left much behind, and some graves remained untouched. Others contained parts of skeletons as well as small grave goods such as bits of pottery, wood coffins, and, occasionally, fragments of grave markers. In one robbers' pit, Stevens and her colleagues found a carved stone grave marker depicting a young Amarna couple seated side by side in an intimate, loving embrace. The marker clearly shows a private couple, says Stevens, and yet, "it strikes you like an image of the royal family." Perhaps, the archaeologist adds, the family of the deceased saw comparable images of the king and queen in an artist's workshop in Amarna and commissioned something similar.

Yet even if people in Amarna sometimes imitated the pharaoh's official art, they were less anxious to embrace his new religion. Little reference to the Aten has been found on funerary goods, and it seems that Amarna's residents preferred to invoke the old gods. Kemp's team has found some skeletons in the South Tombs cemetery adorned with pendants shaped like Bes, an old bandy-legged deity thought to protect homes and ward off evil spirits, and Taweret, the so-called "Mistress

of the Birth House," a favorite of women in labor. In one burial, excavators found a wooden coffin painted with a figure that depicts either Anubis or Duamutef, two funerary gods Akhenaten had rejected.

Despite these small luxuries bestowed in death, and a perhaps unexpected level of religious freedom, however, the lives of commoners in Amarna were far from easy, says bioarchaeologist Jerome Rose of the University of Arkansas. With a team of students and colleagues, Rose is examining the skeletal remains of nearly 400 individuals exhumed from the South Tombs cemetery. To date, the team has studied and photographed the bones of 350 people, and the results are surprising. Says Rose, "I thought that people should be relatively well-nourished, given the abundance



Skeletons excavated in Amarna's South Tombs cemetery reveal that the city was never actually a place of plenty for most of its inhabitants, the majority of whom show signs of severe malnutrition.

of food resources illustrated in the tombs."

Amarna's working class, however, was shorter in stature than almost all other ancient Egyptian populations analyzed by bioarchaeologists using comparable data. Indeed, a 2013 study by Rose and his colleagues showed that Amarna's

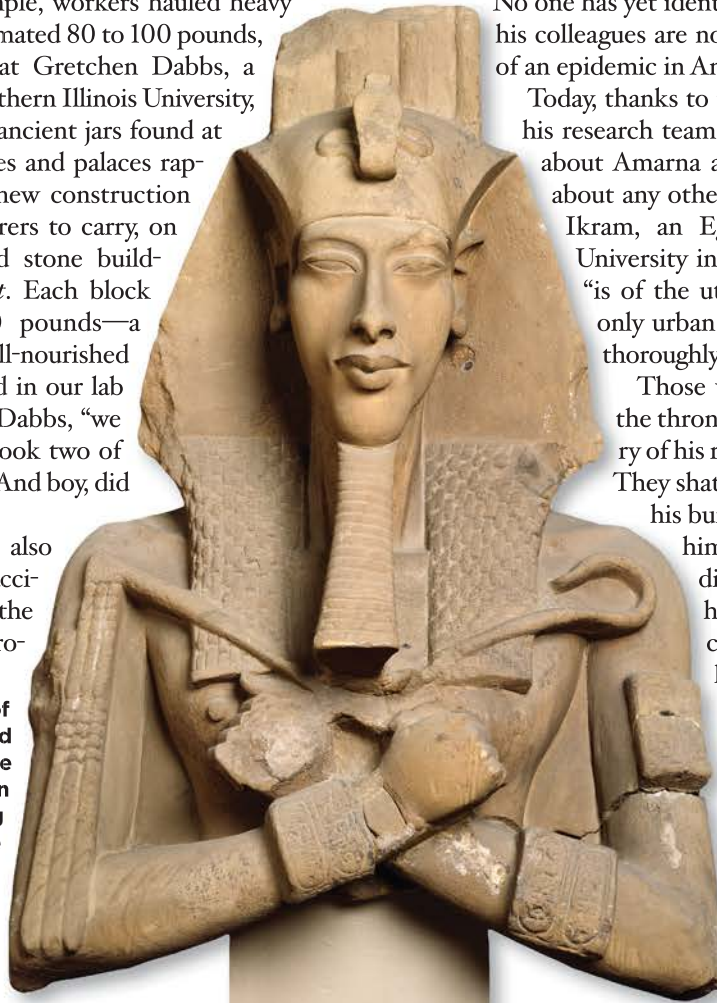


women averaged just a little more than five feet tall, the men only slightly more than five foot four. The data suggests that most inhabitants consumed diets so poor in protein and vital nutrients that they were unable to reach full height. On the skeletons of both children and adults, Rose and his students discerned telltale traces left by iron-deficiency anemia, as well as lesions caused by scurvy, a disease resulting from vitamin C deficiency. Although the people of Amarna lived in a royal capital receiving taxes from across Egypt, they seemed to enjoy relatively little of the bounty.

There were other indications of hardship, too. Some 77 percent of the adults suffered from osteoarthritis resulting from the heavy manual labor needed in a royal capital. To keep gardens green, for example, workers hauled heavy pots of water weighing an estimated 80 to 100 pounds, according to calculations that Gretchen Dabbs, a physical anthropologist at Southern Illinois University, performed on the volume of ancient jars found at Amarna. And to build temples and palaces rapidly, overseers introduced a new construction technique that required laborers to carry, on one shoulder, a standardized stone building block known as a *talatat*. Each block weighed approximately 150 pounds—a heavy burden even for a well-nourished man today. “When we arrived in our lab space two years ago,” recalls Dabbs, “we found a *talatat* block and it took two of our male students to move it. And boy, did they struggle.”

Rose and his colleagues also found evidence of frequent accidents. Nearly 67 percent of the adults suffered at least one bro-

Although many images of Akhenaten were destroyed after his death, this sculpture from the Temple of Aten in Karnak remains as a striking reminder of his innovative style and revolutionary reign.



Analysis by bioarchaeologists is revealing that Amarna's inhabitants led lives of grinding toil. The skeletons show a high incidence of osteoarthritis and a pattern of broken bones characteristic of performing heavy manual labor.

ken bone, including both long bone injuries and compression fractures of the vertebrae. “It’s the kind of damage that you would see associated with industrial work,” says Rose. “It wasn’t just the building of the city, it was also the running of the city.”

Yet accidents, severe malnutrition, and physical exhaustion may not have been the worst of it—a deadly infectious disease may have spread through the city, too. In one part of the cemetery, excavators encountered a series of disorderly, almost jumbled burials, with some graves containing multiple individuals. Mourners, it seemed, had little time or energy to devote to normal burial practices. Moreover, many of the dead interred in that part of the cemetery—in fact, a higher percentage than in other areas—were children.

“Akhenaten was using a place that no one had lived in before,” says Rose, “so I figured there should be fewer infectious diseases.” Instead, these burials raise the possibility of an ancient epidemic, an idea that is also supported by slightly later records from the Hittite Empire to the north. Between 1321 and 1295 B.C., Hittite scribes recorded on tablets the prayers of their king, Mursili II, whose father and brother had both died earlier from a fearsome disease that infected the Hittite population. In his prayers, Mursili traced the source of the infection to Egyptian soldiers taken prisoner on the battlefield.

No one has yet identified the disease, but Rose and his colleagues are now investigating the possibility of an epidemic in Amarna.

Today, thanks to the tireless work of Kemp and his research team, Egyptologists know far more about Amarna and its workings than they do about any other ancient Egyptian city. Salima Ikram, an Egyptologist at the American University in Cairo, says that Kemp’s work “is of the utmost importance, as it is the only urban site from ancient Egypt to be thoroughly and scientifically investigated.”

Those who succeeded Akhenaten on the throne attempted to erase all memory of his rule, his religion, and his capital. They shattered his statues, disassembled his buildings, and erased mentions of him in historic records. But, once discovered, Amarna remains as his enduring monument. The city may not have been what it long seemed, a place of beauty and prosperity, but the gritty reality is proving far more compelling. ■

Heather Pringle is a contributing editor at ARCHAEOLOGY.

PIECES OF THE **SONG DYNASTY**

Investigating a hub of Chinese porcelain manufacture

by LAUREN HILGERS

Jingdezhen has been the source of China's finest porcelain for hundreds of years. Countless pieces have been found in and along the Chang River there.



AMONG CHINA'S ARCHAEOLOGICAL treasures, ceramic sherds, because of their ubiquity, tend to be an afterthought. They pepper farmland and construction sites all over the country, but can hardly compete for attention with the complete bowls, vases, and jugs found every day, to say nothing of entire terracotta armies.

But a stretch of river in southeastern China, long known as one of the country's main production sites for the fine white ceramic called porcelain, is now reminding us how important these often-overlooked artifacts can be. In summer 2013, a small international team of archaeologists went looking for broken ceramics in the Chang River in Jingdezhen in southeastern China. The local cultural bureau had given the group, including researchers from France, Taiwan, and the United States, permission to remove artifacts for examination, but not actively dig. For eight days, the team, led by independent French archaeologist Damien Leloup, dove to the river bottom to retrieve countless ceramics, mostly dating from the Song Dynasty (A.D. 960–1279). There were fragments, near-complete pieces, and even bowls and ewers still within their saggars—protective ceramic boxes used to shield porcelain from the smoke, flame, and gases inside a raging kiln.

This stretch of river was once home to a series of Song Dynasty kilns, and today portions of the shoreline appear to be made almost exclusively of hunks of broken pottery. The river itself holds even more distinctive pieces, including porcelain. “[The pieces in the river] could have been dropped, or, more likely, this could have been a dumping site for wares with slight imperfections,” says Leloup. Even small imperfections—improper temperatures in the kiln can cause porcelain to melt—might have landed a piece in the river or the kiln's dump.

Taken together, these artifacts provide a window into the long history of ceramic production in dynastic China, illustrate some of the innovations that took place at the time, and reflect the extraordinary craftsmanship employed in the manufacture of ceramics—especially porcelain.

FOR MORE THAN 1,700 YEARS—since the earliest days of porcelain-making during the Eastern Han Dynasty (A.D. 196–220)—Jingdezhen has produced China's finest ceramics. In fact, the kaolin clay used to make porcelain all over the world is named for a nearby village. Porcelain, which originated in China, is made by firing a mixture of kaolin and other materials at a high temperature to create a



While imperial porcelain has been well studied, many of the utilitarian wares produced at Jingdezhen have been overlooked.

strong, nonporous, glassy, translucent ceramic. Techniques and style developed over time, and by the period of the Song Dynasty, China's craft and artistry in porcelain production were exceptional. The Ming Dynasty (1368–1844) brought Chinese porcelain to Europe, where the blue-and-white versions became particularly coveted and valuable.

It was access to excellent raw materials, including both kaolin and quality stone, and its location on the Chang River, which connects to the Yangtze through Lake Poyang, that made Jingdezhen such a ceramics hub. The porcelain produced there graced the halls of imperial palaces throughout multiple dynasties and was part of a booming international trade. In 1712, when a Jesuit priest named Père d'Entrecolles arrived in Jingdezhen, he described a town overwhelmed by furnaces. "The whirling flames and smoke, which rise in differ-



Pieces found by the team at Jingdezhen come in a variety of shapes and sizes. The area might have been a dumping ground for imperfect or broken items.



Much of the porcelain in the Chang River is still in saggars, or boxes that protected porcelain during firing (foreground). Jingdezhen once hosted as many as 300 kilns, and portions of the riverbank seem to be made up entirely of broken pottery (background).

ent places, make the approach to Jingdezhen remarkable for its extent, depth, and shape," he wrote in a letter. "During a night entrance, one thinks that the whole city is on fire, or that it is one large furnace with many vent holes." Even today the city

remains China's "Porcelain Capital."

During the Song Dynasty, there were 137 official kiln sites in Jingdezhen, and some accounts put the total number above 300. The ceramicists there excelled in making Qingbai-style ceramics, prized for their delicate bluish-white coloring. These pieces were so fine that they were considered appropriate tribute when visiting the imperial court. Production at a handful of elite kilns fell under the supervision of a government official sent from the imperial "porcelain office." With this system in place, production orders would be sent from imperial courts, and the Jingdezhen kilns

would gather their top ceramicists to fulfill them.

"Most people in China are interested in the imperial kilns," says Leloup. "There were so many kilns in Jingdezhen that the sites with imperial seals have taken priority [for excavation and study]." The Jingdezhen Institute of Ceramic Archaeology has studied a number of these sites and revealed much about the chronology of porcelain styles and many details about its manufacture there. Kilns whose wares were intended for a less illustrious population, such as the examples the team recently retrieved from the river, have largely been left either undisturbed or unstudied prior to development for commercial or residential purposes. This is not uncommon in modern China. In cities with long imperial histories, archaeologists are regularly called in to construction sites to evaluate and document archaeological finds. But there is so much material that heritage authorities are forced to make difficult decisions about which sites are excavated or studied and which are not. This is the case on the Chang River site, where some of the Song Dynasty ceramics appear to have been deposited in the river more recently. "Some buried pieces may have been pushed into the river by trucks," Leloup says.

Among the most important pieces that Leloup and his team uncovered are ones that differ in design from the popular styles of the time. Song Dynasty ceramicists were noted innovators, but official oversight produced ceramics that followed specific guidelines. Non-imperial kilns were more likely to stray from these strictures and respond to demands from trade partners—Song China did brisk business with the Middle East—or domestic collectors. Leloup says they have found at least 14 different, unconventional ewer shapes. "Most kiln sites at the time in China would not dare, or even think of, changing their successful shapes and products," says Leloup. "But Jingdezhen became the most famous of all Chinese porcelain-making sites not just because of the excellent quality of the earth ingredients used to make the porcelain, but also because they were among the first to diversify the shapes and size of their wares." ■

Lauren Hilgers is a freelance writer based in Brooklyn.





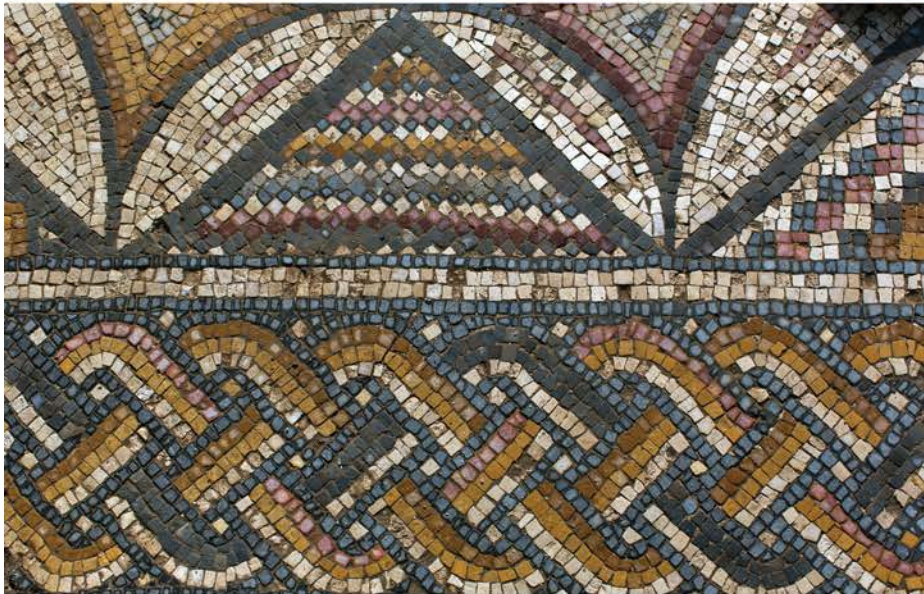
Mosaic Masterpiece

A splendid fifth-century church is evidence of a thriving Byzantine community

by JARRETT A. LOBELL

AT 11 A.M. ON THE 22ND of May 1863, on one of his many trips to the Holy Land, the French explorer and amateur archaeologist Victor Guérin stopped at the small Arab village of Hatta, which he described as surrounded by tobacco plantations and pine, olive, and fig trees. There Guérin saw four column drums lying alongside a well, “three of grey-white marble, and the fourth of a bluish marble,” as well as several other stones which “offered the same ancient appearance.” What Guérin did not know as he continued his journey the next morning was that the columns were once part of a large fifth-century church that has only just been revealed more than a century and a half later.

During recent salvage excavations before the construction of a new neighborhood, archaeologists from the Israel Antiquities Authority (IAA) uncovered a building 98 feet long and 47 feet wide, built of sandstone, limestone, and marble, with magnificent mosaic floors. Though the IAA had excavated some Byzantine (A.D. 330–638) sites in this area, and knew from Guérin as well as a British Mandate map of 1945 that there may have been an ancient building there, nothing significant had been found until they uncovered the church at the beginning of the year. In antiquity, the church would have been on the main road between the important ancient cities of Eleutheropolis and Jerusalem to the east and Ashkelon on the coast to the west. According to project director Daniel Varga, “the building was probably the central church for communities in the area, and perhaps part of a monastery that we have not excavated.” Varga explains that in order to best protect the remains, the mosaics have now been removed for conservation and the site has been reburied. ■





The church's remarkable mosaic floors display an array of animals, plants and fruits, inscriptions, religious symbols, and geometric patterns.

Clockwise from above:

Amid swirling grape vines, an inscription in Greek names the church's archdeacon, Demetrios, and his fellow deacons, Zurgaios and Hyperechios.

A conservator cleans a mosaic of two birds and a Christogram, a type of monogram of the name of Jesus, which is flanked by an alpha and an omega, the first and last letters of the Greek alphabet, recalling the characterization of Christ as the "beginning and the end."

Geometric patterns made using colorful stones of local origin can be found throughout the more than 1,600 square feet of mosaic flooring.

Unlike synagogue mosaics, church mosaics tend to avoid biblical episodes and instead contain vignettes of rural and natural life, including local and exotic animals such as leopards, zebras, and wild boars.

Jarrett A. Lobell is executive editor at ARCHAEOLOGY.



Chinatown, San Francisco, 1896-1906



America's Chinatowns

Dozens of digs and collections are revealing the culture, diversity, and challenges of the first Chinese Americans

by SAMIR S. PATEL

BEFORE THE CALIFORNIA Gold Rush in the late 1840s, there were perhaps 50 native Chinese people in the United States. Just a few decades later, there were more than 100,000, and seemingly every city, town, and remote mining camp in the West had a Chinatown of its own. Chinese immigrants were exotic curiosities, targets of racism and violence—and an essential part of the labor force that settled the West. They left little written history, but dozens upon dozens of archaeological sites and collections are now enriching our understanding of how the first Chinese Americans negotiated life in a strange and sometimes hostile land.

Most of the nineteenth-century Chinese immigrants came from an area the size of Rhode Island—Taishan County, in the southern province of Guangdong, which had suffered the dual indignities of the Opium Wars and the Taiping Rebellion in the 1840s and 1850s. The opening of trade relations between China and the United States, and the discovery of gold in California, spurred the first surge of immigration. In 1852 alone, more than 20,000 Chinese people passed through San Francisco's Golden Gate. Many found relative safety, comfort, and job opportunities in Chinatowns, which grew first in the cities and then appeared on the frontier as Chinese laborers pursued work in railroad construction, mining, lumber, agriculture, and other industries. Their population in the United

States declined following the Chinese Exclusion Act of 1882, which prohibited new immigration, and would not rebound until the restrictions were lifted 60 years later, starting a second wave of Chinese immigration that has since brought their numbers north of three million.

Archaeological investigation of Chinatowns and Chinese neighborhoods began in the 1970s, with digs at sites such as Ventura, California, and Lovelock, Nevada. There were more in the 1980s and 1990s: Sacramento, San Francisco, Oakland, Los Angeles, San Jose, and others. But most of this early work was cultural resource management—digs related to construction or roadwork—and generated little analysis or scholarship.

Today, Chinese-American archaeology is changing, with new digs, the rediscovery of old collections, and a push to bring researchers together to share findings. Archaeologists and historians have begun working closely with historical societies and descendant communities, and even collaborating with colleagues in southern China. The picture emerging is of a complex, diverse community that held on to some traditions, selectively adopted aspects of Euro-American culture, and tried to make the most of opportunities.

"One of the things that archaeology is doing for all of Chinese America is giving us a greater understanding of what transpired among people who left no or very little written record," says Sue Fawn Chung, a historian at the University of Nevada, Las Vegas, who specializes in Chinese-American history.

Culture

Nineteenth-century Chinese immigrants were usually regarded with deep suspicion by Euro-Americans. Their goods were considered strange, their women immoral, their neighborhoods dangerous, their food unsavory. The first studies of their material culture in the 1970s and 1980s described early Chinatowns as insular, traditional, and resistant to change. These early archaeological studies focused on the idea, long abandoned by historians, of acculturation—that is, how much the Chinese were assimilating American culture. “I think that the way we thought about cultural change was a little narrow,” says Barbara Voss, a Stanford University archaeologist who leads the study of the Market Street Chinatown collection in San Jose.

Archaeological thinking has since evolved, and the relationship between Chinese immigrants and American culture is now known to be much more than a one-way street toward



Arson fire, Market Street Chinatown, San Jose, 1887

“Americanness.” Chinese-American culture was shaped by tradition, connections to mainland China, institutional segregation and racism, local circumstances, and a complex process of adaptation and selective accommodation.

Examples of this complexity come from a number of projects. Market Street Chinatown in San Jose was destroyed in an arson fire in 1887. Anti-Chinese sentiment was high, and arson, beatings, and even murder were not uncommon. The burn layer preserved the site, which was excavated about a hundred years later. The collection, however, was packed away, only to be rediscovered in a storage room in 2001 by archaeologist Rebecca Allen of private cultural-management firm ESA—all 451 boxes of it—who brought it to Voss’ attention. Analysis of the collection (some 50,000 artifacts to date) continues, but has so far revealed a community that was hardly isolated. A rich mixture of Asian, British, and American household goods shows that though it was an ethnic neighborhood, its borders were porous, and its popula-



Ceramics, Market Street Chinatown, San Jose

tion diverse, mobile, and adaptable. “It’s very clear culture is not a zero-sum game,” Voss says. “People were not choosing between tradition and innovation.”

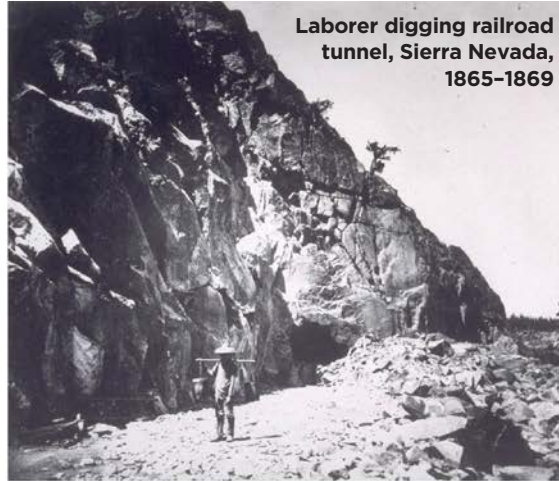
During excavations in Sacramento, Adrian Praetzelis of Sonoma State University found a specific instance of how the Chinese selectively, even strategically, adopted American goods. Chinese agents—bilingual businessmen who worked with Americans and helped Chinese laborers find work—were common in urban Chinatowns. In what was once the home of an agent, Praetzelis found both Chinese ceramics and British pieces, especially large serving dishes. According to historical sources, agents held open houses with local businessmen and politicians to show that Chinatown was a reliable place to conduct business. At these banquets, Chinese food was served but—as Praetzelis’ finds show—probably on British servingware. The British ceramics were a nod to guests rather than a sign of assimilation. “It had little or nothing to do with culture change itself,” says Praetzelis. “It had to do with intelligent manipulation of what could be a difficult, even dangerous, situation.” Interestingly, the ceramics include pieces of willow pattern, an iconic nineteenth-century British print that itself had been adapted from a native Chinese pattern and mass-produced in England.

Euro-American culture was not the only influence on Chinese communities. At work camps, laborers from different ethnic backgrounds often lived and worked side by side. At a cannery site in Canada, for example, Doug Ross of Simon Fraser University is examining the differences and similarities between Chinese and Japanese work camps. Furthermore, many neighborhoods considered “Chinatowns” were also home to other marginalized or segregated populations—such as other Asians, African Americans, Native Americans, Portuguese and German immigrants, and even white prostitutes—who found safety, community, a functional economy, and less discrimination or judgment.

Labor



Summit Pass, Sierra Nevada



Laborer digging railroad tunnel, Sierra Nevada, 1865-1869

The Chinese first migrated to the West for the same reason most everyone else did: a shot at “Gold Mountain.” As their reputation for hard work (at low wages) grew, more Chinese came and found jobs on the railroads. Though Chinese immigrants worked in other professions, at the peak of railroad construction in the 1860s, up to 90 percent of the labor force was Chinese. They dug trenches and tunnels, built grades, and laid track from Utah to California, through the same country that had famously claimed the lives of the pioneer Donner Party just 20 years earlier. “The Transcontinental Railroad, arguably one of the most significant engineering feats in North American history, was constructed largely by Chinese workers,” says Stanford’s Voss. “It’s a dramatic example of the role that workers play in building countries and societies.”

Many of the camps that housed Chinese railroad workers were temporary and have long been forgotten. But between 1865 and 1869, the Central Pacific Railroad pushed through Donner Pass in the High Sierras. At 7,000 feet, sometimes with up to 40 feet of snow pack, laborers, many of whom were Chinese, dug seven tunnels through solid rock. The work took four years, so some encampments there were more established and longer-lived than those elsewhere. John Molenda, a graduate student at Columbia University, has surveyed of a number of these camps, and Rebecca

Allen and Scott Baxter of ESA studied one at Summit Pass in detail. “We wanted to know what they were bringing there to make themselves feel like they were at home,” says Allen.

Because tools and personal items probably traveled with the workers, the remains at these sites include some stone foundations and an array of potsherds, gambling pieces, and opium paraphernalia (perhaps used to treat injuries, aches, and pains). At the Summit site, Allen found the remains of a large roasting hearth, evidence of communal meals that may have helped build social bonds. Gambling pieces served a similar purpose, says Molenda. These men were away from their families or the relative comforts of a Chinatown for years at a time, and gambling was a way to create community. Besides, he adds, they wouldn’t have had much to lose: Chinese immigrants sent around two-thirds of their wages back home. According to Molenda, this sense of community, combined with a distinctly Chinese sense of duty and sacrifice, made them particularly effective and sought-after laborers. “They did as much as they could to make themselves as comfortable as possible within the constraint that what they were doing was not about

having fun,” says Molenda.

They were so effective, in fact, that many American investors moved specific work gangs onto other projects after the completion of the railroad, and even sometimes protected them during surges of anti-Chinese sentiment. To the capitalists, the Chinese were a

cheap, easily controlled labor force. To keep them that way, most investors supported anti-Chinese legislation, which thwarted any social mobility.

Much of this recent research is coming together in the Chinese Railroad Workers in North America Project at Stanford University, a cross-disciplinary effort involving historians, archaeologists, and other scholars. The project aims to create a digital archive of research materials, organize events and conferences, and make connections with institutions in southern China to unify the study of the railroad workers. According to Voss, archaeological director of the project, it is an opportunity for archaeologists in the field to examine new methods for studying a mobile, rapidly moving population. “This is a good project for archaeology,” she says. “This forces us to think differently about our data.”

Gambling pieces, Chinese work camp, British Columbia



Health

Life would have been hard for most Chinese immigrants in the American West, especially itinerant workers. Ryan Harrod of the University of Alaska Anchorage recently excavated 13 sets of skeletal remains from a Chinese graveyard at Carlin, Nevada, the district terminus for the Central Pacific Railroad. Such graves are rare, as remains were usually sent home to China. The individuals buried at Carlin had several signs of poor nutrition. They also showed evidence of years of grueling manual labor: highly developed muscle attachment points and joints with signs of deterioration. There was also blunt force trauma, from broken bones to a fatal head wound, which reflects the constant threat of physical violence they faced. "In general, just being Chinese at a time when there's anti-Chinese sentiment in a fairly rural community in the West probably wasn't that good a situation," says Harrod.

Daily life might have been better in more established Chinatowns, but the Chinese still found themselves disadvantaged. "They were often denied access to public healthcare

facilities and denied treatment by doctors," says archaeologist Sarah Heffner, who examined seven collections of Chinese-American healthcare-related artifacts for her dissertation at the University of Nevada, Reno. In the collection from Lovelock, Nevada, at the Nevada State Museum, Heffner found more than 100 health-related artifacts. She found herbal materials used in

Chinese medicine, such as turtle carapace, cuttlefish cuttlebones, snake bones, and bobcat bones (perhaps a local substitution for tiger bones, prized in traditional Chinese medicine). There was also packaged Chinese medicine.

One embossed single-dose vial had the name of a drugstore in Hong Kong and specified that it contained "deer's tail," "testes and penis of ursine seal," and "tonify the kidney pills"—all associated with kidney health and treatment of impotence or premature ejaculation. The paper wrapper from another container was labeled "bao ji

Single-dose medicine vial



"Pills of relief" wrapper

wan," or "pills of relief," and had a written warning that any attempts to copy the contents would result in a curse on the counterfeiter's family.



Snake bones

In the Lovelock collection Heffner also saw Euro-American medicines, primarily bottles containing popular patent medicines such as Sloan's Family Liniment and Perry Davis's Vegetable Painkiller. Like the packaged Chinese medicines, these purported cure-alls

Food

It doesn't take a connoisseur to know that much of the Chinese food sold in the United States today is hardly the same as traditional Chinese cuisine. We can think of it as Chinese-American food, and its roots likely lie in the kitchens, restaurants, and hearths of nineteenth-century Chinatowns in the West.

Since most of the early Chinese immigrants came from southern China, their cuisine was based on the Cantonese food cooked in Guangdong, mostly rice, vegetables, and pork or fish. The American populace vilified Chinese food as smelly, suspect, and unhealthy. The Chinese cooked their

traditional food using both imported ingredients and local substitutes, and catered to local tastes when they had to. "It is fairly similar to other immigrant food situations around the world and in the past," says Ryan Kennedy, a graduate student at Indiana University who is studying the subject.

The collection from Market Street Chinatown in San Jose includes around 140 soil samples taken from what were probably food storage areas. Stanford's Voss recently sent samples to Linda Scott-Cummings of the PaleoResearch Institute in Golden, Colorado, to have them analyzed for seeds, pollen, and phytoliths (small pieces of silica

that form in plant tissues). There was evidence of rice, as one might expect, but there were also signs of wheat, barley, and millet, which might have been used to make northern Chinese-style noodles.

Scott-Cummings also found evidence of a huge range of vegetables—some familiar to Chinese cooking (bitter melon and jujube, also known as a Chinese date) and some new (gooseberry, walnut), as well as indications of interaction between San Jose's Chinese and Hispanic populations (corn and tomatillo). Among the standout finds was pollen from agave. Young agave buds look much like lily buds, which are

would have been cheap and practical. “The list of ailments they treat goes on and on and on,” says Heffner. The use of both traditional Chinese treatments and American patent medicines is indicative of a lack of access to more formal healthcare. Not to mention that American patent medicines had a certain popularity because they could be up to 40 percent alcohol and exempt from taxes on spirits.

Single-dose vials like the ones that Heffner studied are common at Chinese sites in the West. Ray von Wandruszka, a chemist from the University of Idaho, is the first to look inside such vials. “Some still had stuff in them,” he says. One vial from Market Street Chinatown in San Jose contained a bright red residue that turned out to be cinnabar, a sulfide of mercury used to treat infections in Chinese medicine. “It’s really toxic, by the way,” von Wandruszka says. Another contained magnetite, used to make an elixir to “anchor and calm the spirit—relieve restlessness, palpitations, insomnia, tremors—improve hearing and vision—combat chronic asthma,” according to a modern sales pitch. These treatments are known as “stone drugs,” pure or mixed mineral powders, and many are still available online today.

used in traditional Chinese dishes. In fact, Scott-Cummings has observed pollen from other members of the lily family at Chinese sites in Wyoming and Montana, suggesting that local populations often sought out substitutions for this cherished food item. “When you’re living on a different continent, you get whatever you can,” she says.

At other sites, archaeologists have found that the Chinese ate much more beef in the United States than they would have in China, where cows were too valuable as beasts of burden to be eaten. “It would have been a departure from what they were eating in China,” says Kennedy. “People are trying new things.”

Women



Woman in Chinatown, San Francisco, 1896–1906

In the 1850s and 1860s, as much as 90 percent of the population of California—not including Native Americans—was male. This imbalance was temporary for Euro-Americans, but not for the Chinese. Most of the immigrants were men, and American employers preferred an all-male workforce. Anti-Chinese sentiment and the legislation that grew from it exacerbated the gender problem. The Page Act of 1875 outlawed the immigration of “undesirables,” and because Chinese women were stereotyped as prostitutes, it effectively barred them from entry. In addition, the Chinese Exclusion Act of 1882 prevented Chinese men from sponsoring the immi-

gration of their wives and families. It was, essentially, American policy that Chinese men not produce natural-born Chinese-Americans. “This was really an American government policy enforced upon the Chinese to create an unnatural bachelor society,” says Chung, the University of Nevada, Las Vegas, historian.

By some estimates, in 1890, Chinese men outnumbered Chinese women by 28 to one in the United States. There were never more than 5,000 Chinese women in the country well into the twentieth century, according to official records, though historians believe they were undercounted by the census. “Generally your husband hid you in the house,” Chung says. There are accounts of some women who had a surprising amount of freedom and success, but their stories seem to be outnumbered by tales of prostitutes, Chinese women probably forced, coerced, or trafficked into sex work. “[Prostitutes] are never absent from the scene,” says Chung. “But we really don’t have any way to gauge the numbers.”

There are few artifacts found at Chinese sites in the West that can be definitively tied to women, though their presence can be inferred from toys that are sometimes found in Chinese neighborhoods, even remote ones. The right site to establish the archaeological signature of nineteenth-century Chinese women has yet to be found.

Chinese men no doubt felt this absence. California was among the states that outlawed marriage between whites and non-whites, so it is no surprise that the Chinese and other marginalized groups, such as Native Americans, might find common ground. At Mono Mills, California, which is being studied by Charlotte Sunseri of San Jose State University, the Chinese laborer community lived side by side with a community of Paiute people. The two groups definitely interacted—Sunseri found Chinese ceramics in Paiute households, while pine nuts and obsidian blades turned up in Chinese homes. It is possible they may even have cohabitated. There are, in fact, historical records of Chinese-Paiute marriages. “They each had their own marginalization and this was a form of agency,” says Sunseri.

Even after the Chinese Exclusion Act was repealed in 1943, setting off a second wave of Chinese immigration, it would be another 20 years or more before the gender ratio normalized among Chinese Americans. ■

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In 1834, an American expedition to the Plains encountered a Comanche chief wielding a white buffalo skin as a flag of truce, immortalized in this painting by George Catlin. The archaeological record of Comanche life has only recently been explored.

Searching for the Comanche Empire

In a deep gorge in New Mexico, archaeologists have discovered a unique site that helps tell the story of a nomadic confederacy's rise to power in the heart of North America

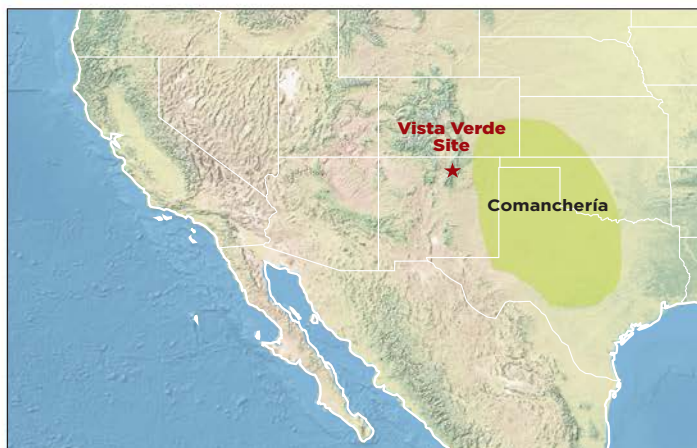
by ERIC A. POWELL

THEY CALLED THEMSELVES NUMUNU, “the people,” and for centuries they had been hunter-gatherers living in small camps in the Rocky Mountains. But sometime before 1701, when they were first documented by the French on a map of the High Plains, the Numunu left the mountains and encountered horses, possibly trading for them with their linguistic cousins, the Ute, or the Pueblo people of northern New Mexico. By the mid-eighteenth century, they were known as Comanche, a name derived from the Ute word for “anyone who wants to fight me all the time,” and, on the strength of their unparalleled equestrian skills, they were well on their way to being the dominant Indian nation of the American West. Among the most feared mounted warriors in history, the Comanche forged a nomadic culture that served as a model for other Plains Indians. They ranged from Canada all the way to central Mexico, and carved out a homeland that would come to be known as Comanchería, which included much of modern-day Texas, New Mexico, Oklahoma, Kansas, and Colorado, and which endured until the mid-nineteenth century.



Until now, despite the fact that they controlled a vast amount of territory for almost two centuries, and at one point numbered some 40,000 strong, the Comanche have been virtually ignored by archaeologists. “We thought the Comanche had a culture designed to be invisible and to escape detection,” says Barnard College archaeologist Severin Fowles. “If they made camps that they could strike so that no trace remained for the U.S. cavalry to find a few days later, what hope could archaeologists have of finding them more than 200 years later?” But the recent identification of previously unknown panels of rock art at a Comanche encampment in New Mexico’s Rio Grande Gorge is challenging the idea that they left no physical traces behind.

The discovery coincides with the rise of a new generation of historians, who, together with the Comanche them-



In the early eighteenth century, Comanche raiding or trading parties pitched camps in the Vista Verde basin (top) in the Rio Grande Gorge west of Taos, New Mexico. On basalt boulders surrounding the site they scratched numerous scenes depicting raids, battles, and tepees. As depicted on the map, the Comanche ranged across a vast territory that came to be known as Comancheria and encompassed much of modern-day Texas, New Mexico, Oklahoma, Kansas, and Colorado.

selves, are rereading colonial records and putting together a revised account of Comanche history. This new view contradicts the image of the Comanche in the popular imagination, which casts them as the most brutally savage of the Plains Indians, whose relentless raiding stalled the expansion of the U.S. frontier for decades. In the new, more nuanced approach to Comanche history, the “Lords of the Southern Plains” are emerging instead as skilled tacticians and diplomats capable of mustering thousands of warriors at one time to advance their political and economic interests.

“The prevailing view had been that the Comanche at best held off colonial expansion until the frontier rolled over them,” says University of Oxford historian Pekka Hämäläinen. “But when we take a broader, more regional look, we can see Comancheria was a real political and cultural center in the Southwest.” In fact, Hämäläinen

and other scholars say that a fresh look at the history of New Mexico and Texas suggests that, for many decades, the Comanche dominated these Spanish, and later Mexican, provinces. They extracted so many resources through “gifts” and raiding that the European settlements were virtual satellite states of Comanchería, more tied politically and economically to a mobile Native confederacy than to the government in Mexico City.

Looking at history from the perspective of the Numunu, some scholars now argue that the Comanche nation had imperial ambitions, and say that it is no exaggeration to describe their story from the late seventeenth to the nineteenth century as the rise and fall of the Comanche Empire. Oral history and written records help tell most of this story, but the early years of the Comanche, when they first forged their equestrian-based way of life, remain murky. “This crucial period is a black



hole in Comanche history,” says Hämäläinen. “And the fact that the archaeological record is so scant has really crippled historians.”

The newly discovered rock art in New Mexico, found by an archaeological team, led by Severin Fowles, and that now includes Comanche Tribal Historic Preservation Officer Jimmy Arterberry, could help illuminate this largely unknown period.

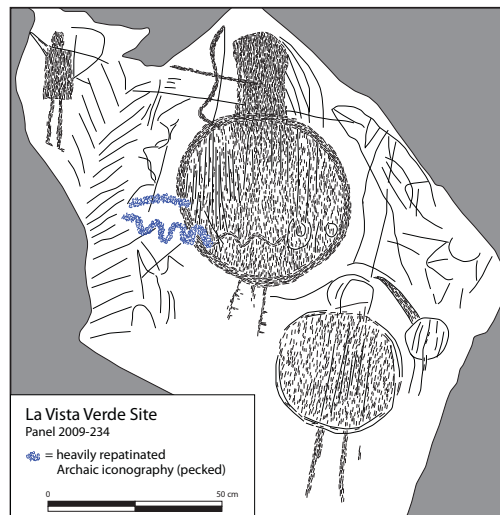
IN 2007, FOWLES AND HIS TEAM began an ambitious survey of the steep slopes of the Rio Grande Gorge, a deep canyon west of Taos. No serious archaeological study had been made of the canyon, and Fowles’ interests at the time were centered on exploring how ancient hunter-gatherers living in the Archaic period (7,000 to 2,000 years ago) and the ancestors of today’s Pueblo people used the gorge as a sacred place to carry out rituals. The team discovered numerous

This painting (above) by Catlin depicts a large Comanche encampment, where women processed buffalo hides, an important trade item. Plains Indians often painted hides with realistic narratives of personal exploits, such as this 19th-century example (right), which shows vignettes of skirmishes and horse raids.





Archaic and Pueblo rock shrines and rock art imagery that both Archaic and ancestral Puebloan people painstakingly pecked with stone tools into basalt boulders throughout the canyon, and that are still unmistakable today. Intent on making a complete record, Fowles instructed his team not to ignore graffiti made by shepherds, cowboys, tourists, and even modern-day gang members, all of whom have left their mark in the canyon over the past 200 years. The team soon found itself recording not just cattle brands cut by cowboys into the cliffs and gang markers sprayed on boulders, but many faint scratches on the rocks that were not easily identifiable as modern or ancient. Many were in the shape of triangles. “We found ourselves asking, who is this weirdo scratching triangles out here?” says Fowles. “It’s humbling to think about it now. We were recording the pecked art so faithfully, and then noting the scratches too, but as an afterthought.” That changed one day during the summer of 2008 when Columbia University undergraduate Annie Dennis took a bathroom break and came back to the group with some exciting news. She had found a boulder that she believed could solve the mystery of the scratches.



Scratched rock art panels at Vista Verde are some of the earliest known archaeological evidence for the Comanche presence in the Southwest. A tracing (left) of scratches on a boulder (above) shows they depict three warriors. The central figure is probably a Comanche who brandishes a shield and exchanges arrows with a smaller warrior to the left. A second shield bearer on the lower right of the boulder might be coming to the larger warrior’s aid.

On the boulder, beneath a layer of nineteenth- and twentieth-century graffiti, were unmistakable depictions of horses and ceremonial hide bags, or parfleches, scratched into its surface in a style that Fowles recognized as belonging to Plains Indians. The mysterious triangles they had already recorded were also on the boulder, and it became obvious that they were depictions of tepees. Now that the archaeologists knew what to look for, they began to find the scratch panels everywhere in the canyon. An especially dense cluster was discovered on rugged basalt ridges that surrounded a large flat basin known as Vista Verde, not far from the boulder Dennis had discovered. These panels, which include depictions of horse herds, battles, ritual objects, and whole encampments of tepees, are in some cases within view from a hiking trail that

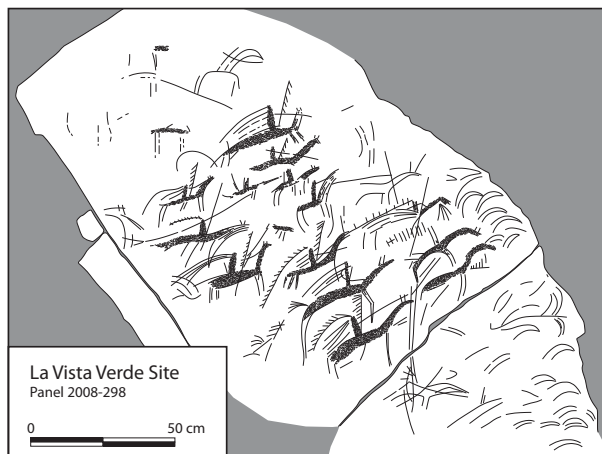
has been used for decades. The barely discernable artwork is a rich archaeological record that no one had ever thought to look for, hidden in plain sight. “It’s hard to find something if you’re not looking for it,” says Fowles.

IMMEDIATELY AFTER DISCOVERING the rock art, Fowles contacted tribes whose ancestors once lived in the northern Rio Grande Valley, including Taos Pueblo, and three formerly nomadic groups with close connections to the area: the Jicarilla Apache, the Ute, and the Comanche. After showing the artwork to representatives of the tribes, including Arterberry, a consensus emerged that the panels had been created by the Comanche. They were known to have both raided settlements and conducted trade in horses, captives, and buffalo meat in northern New Mexico since at least 1706, when Spanish records first mention them as a threat to Taos Pueblo. Many of the Pueblos still preserve the memory of these interactions by celebrating the Comanche Dance, in which they dress as Plains Indians and reenact those fraught encounters. Around Christmas, Hispanic villages in the area long staged a folk play called “Los Comanchos,” which

revolved around conflict between a Hispanic community and a group of raiding Comanche. The village of Alcalde, 25 miles downstream of the Vista Verde site, still holds the play on horseback every winter.

Despite these deep Comanche ties to the area, prior to Fowles’ discovery not a single Comanche site had been recorded in northern New Mexico. But the style of the rock art at the Vista Verde site made Arterberry confident that his ancestors had once pitched their tepees there, either sheltering in the canyon before or after raiding or while en route to the annual trade fairs held at the Pueblos and Hispanic villages. The depictions of tepees, horses, and warriors are all unmistakably in an early style of the Biographic tradition of Plains rock

art, in which the Sioux, Crow, and Blackfeet, among others, used realistic forms to narrate actual



A tracing (left) of a large scratch art panel (below) at Vista Verde reveals that it depicts a dozen warriors with lances and long ropes charging on horseback after arced lines that represent dozens of horses. The art probably tells the story of a specific horse raid, which would have been an important source of wealth and prestige for the Comanche who created the panel.



events, such as raids or buffalo hunts. The Comanche were known to have made similar, isolated images in sandstone, but the sheer mass of depictions at Vista Verde is unprecedented. “I’ve seen lots of individual Comanche rock art panels,” says Arterberry. “But this is unique, to see them together. Not only is all our history here, but walking through the site is like walking through a timeline.”

LAST AUGUST, ARTERBERRY AND FOWLES visited Vista Verde to share their thoughts on the site, which sits about halfway up the side of the gorge and overlooks the confluence of the Rio Grande and the Rio de Taos Pueblo. The Comanche are known to have preferred camping where two rivers came together, which they believed had strategic value. With several Spanish settlements less than half a day’s ride from Vista Verde, it was an ideal location from which to launch raids or trading missions.

Above the Vista Verde basin are two stone alignments, which Fowles notes could have served as blinds for Comanche warriors guarding the main camp on the flat ground below. Two boulders nearby, including the one discovered by Dennis, are covered with images of parfleches, which had sacred significance to the Comanche. On one boulder, there is an almost life-size depiction of a horse carrying a rider who bears a shield. Short scratches emanating from the horse’s mouth seem to represent the animal’s breath, perhaps condensing on a cold

winter day. The images are difficult to make out in the direct sun, but when a cloud passes overhead, the horse seems to suddenly leap off the rock surface. Recording the images on the panels is painstaking work, since each individual scratch has to be traced across the rock surface. “It’s almost as if we have to reenact the gestures of the Comanche artist who made it three centuries ago,” Fowles explains. Often, it’s not clear what the image depicts until the tracing is completed and digitized.

Sheltered as it is by basalt ridges, the Vista Verde basin would have been an ideal place to pitch a large group of tepees. Fowles’ team has found several tepee rings here, and remote sensing shows still more lying just a few inches beneath the surface. The team failed to find any clear evidence of hearths, meaning Vista Verde was at least at times a “cold camp,” where no fires were made for fear the enemy would see them. On the boulders around the edge of the basin, Comanche warriors scratched numerous battle scenes filled with a number of telling details. One depicts two warriors shooting arrows at one another. One of the combatants is small and wields a simple bow. The other, presumably a Comanche warrior, is much bigger and brandishes a larger bow. Next to him a “tally” of arrows suggests he shot many more arrows than his smaller opponent. He is also wearing unusual headgear that Arterberry thinks is a Catholic missionary hood, which Comanche warriors were known to steal in raids and then wear into battle. A stylized bear is

depicted rising behind the warrior, along with a shield bearer, and both seem to be coming to the warrior’s aid. The shield bearer wears a war bonnet that to Arterberry seems to be in the style of the Ute, for whom the bear was sacred. In the early eighteenth century, the Ute and Comanche were close allies. If the third figure is Ute, the scene could memorialize an important diplomatic relationship that helped enable the Comanche rise to power in the Southwest.

Not far from this panel, the team’s remote sensing devices picked up an unusual horseshoe-shaped anomaly. Ute tribal members identified it as a clearing for the Bear Dance, an important Ute ritual. In the early eighteenth century, before their relationship with the Comanche collapsed and the two groups became bitter rivals, the Ute may have camped here with

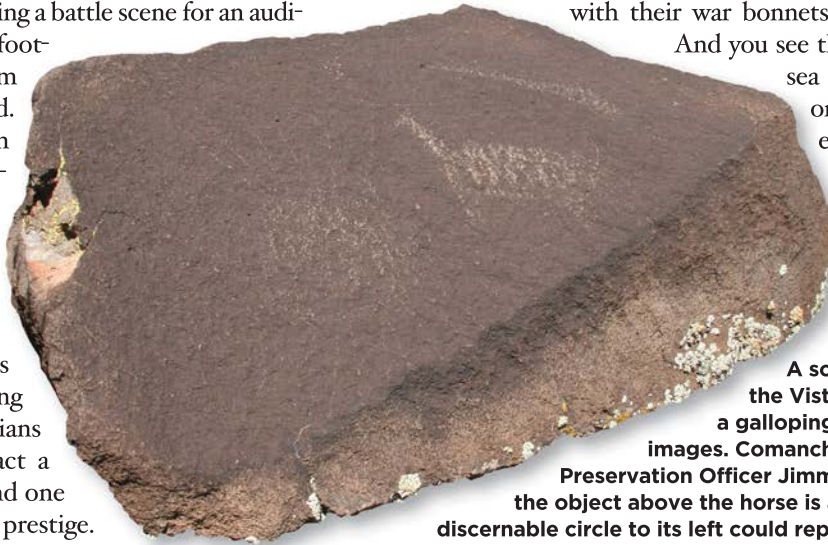


The horse was central to the Comanche way of life, and the tribe was renowned for its feats of horsemanship, captured here in a painting by Catlin.

their then-allies. “It’s interesting that there are no images of Europeans at the site,” says Fowles. “The Comanche are depicting a world where Indian-to-Indian relations are still much more important than relationships with Europeans.” So far, the team has not found any warriors at Vista Verde depicted with guns, which has led them to date the rock art to before the 1740s, when the Comanche began to acquire rifles from the French.

Across the basin, more faintly scratched images depicting raids in progress have been found on boulders. Most of these chaotic scenes are confusing to anyone trying to make them out today. Fowles thinks that is because, unlike most rock art, these panels were not intended as permanent works, but were created by warriors narrating a battle scene for an audience, much the way a football coach might diagram a play on a chalkboard. Some of the warriors on the panels have meandering lines coming from them that touch other combatants. These could be depictions of “counting coup,” or the act of touching your opponents in battle without harming them, which Plains Indians considered the bravest act a warrior could commit, and one that would win him much prestige. Fowles and Arterberry believe that Comanche told their stories of bravery and scratched the scenes into the rock before an audience of fellow warriors so their deeds would be publicly acknowledged. The desire to boost individual prestige through battle was one of the engines of Comanche society; the only way some young bachelors could expect to win a wife. “Fowles and his team are showing how this rock art illuminates the social dynamics and the ambition of the Comanche in this early period,” says Hämäläinen. “It’s very exciting.”

The raiding depicted in the scratch panels at Vista Verde helped shape not only Comanche history, but the history of North America as a whole. In the early nineteenth century, the descendants of the warriors who camped at Vista Verde made military forays into northern Mexico’s mining districts, once coming within 200 miles of Mexico City itself. These expeditions, along with raids by other Native groups, sapped the military strength of northern Mexico. Because of the desolation caused by these campaigns, University of California, Berkeley, historian Brian DeLay calls this conflict “The War of a Thousand Deserts,” and argues that it left Mexico vulnerable to the American invasion of 1847, which in turn led to the U.S. annexation of the Southwest and California. “The Indian raiding certainly influenced the outcome of the U.S.-Mexican War,” says DeLay. “I think you have to think of the Comanche as important geopolitical players.”



NOT FAR FROM THE CAMPSITE at Vista Verde is a large stretch of ground bounded by cliffs that would have made a perfect natural corral. A spectacular panel scratched on a boulder here depicts a horse raid in progress, perhaps one of the raids that periodically left the colony of New Mexico nearly bereft of mounts. On the panel, a dozen warriors with war bonnets streaming behind them charge on horseback after dozens of arced lines that are stylized depictions of horses. It is Comanche history recorded by the Comanche themselves.

For Arterberry, who is also an artist, it’s the most important panel at the site. “I love this sea of horses,” he says. “As an artist, I see a change in tradition from two-dimensional art to this almost abstract style. You can clearly see the warriors on horses, with their war bonnets, lances, and long ropes.

And you see them in juxtaposition to a sea of abstract horses, each one drawn distinctively and each one conveying a multitude of horses. It’s very contemporary.”

Depictions of single horses, of warriors riding horses into battle,

A scratch panel found near the Vista Verde site features a galloping horse and two other images. Comanche Tribal Historic Preservation Officer Jimmy Arterberry believes the object above the horse is a comet. A barely discernable circle to its left could represent the sun.

and even of stylized tepees riding horses, abound at Vista Verde. In 1875, when the U.S. cavalry surprised and routed the last nomadic Comanche band in Texas’ Palo Duro Canyon, the first thing the soldiers did was slaughter the group’s horses. Only with their horses gone did the band finally surrender.

In summer 2013, Fowles showed Arterberry a new panel the team discovered not far from the Vista Verde site. Fowles and his group thought it depicted a horse and some other objects, but were uncertain. Upon seeing the panel, Arterberry knew instantly that it showed a galloping horse and a comet in the sky, both heading toward the sun. After looking at historical accounts later, Arterberry would find that in 1680 there was a comet visible by daylight. Looking at the panel in the field, though, he knew that the Comanche who created it was celebrating the connection between the horse, the sun, and a celestial event. “Anthropologists sometimes call us sun worshippers,” says Arterberry. “We’re not, really. We think of the creator as our father, a great deity beyond, who the sun can represent. But to see the sun and the horse and the comet here is very powerful. The creator gave us the horse, and that’s the beginning of the story.” ■

Eric A. Powell is online editor at ARCHAEOLOGY.org.

For more Comanche rock art, go to www.archaeology.org/comanche

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Archaeologists dig in a garden near the house of John Bartram, known as the father of American botany.

City Garden

**The unlikely preservation of thousands of years of history
in a modern urban oasis**

by MARGARET SHAKESPEARE

A short trolley ride southwest from Philadelphia's Center City, on a gentle curve in the Schuylkill River, lies a bucolic spread of land incongruously situated within a shabby urban neighborhood. In the shadow of an eclectic eighteenth-century Georgian-style stone house, there is a sprawling garden and, beyond, a meadow, rolling green hills, woods, mudflats, and wetlands. The urban skyline is distant but visible through clearings in the trees. In the spring, trillium, azaleas, and bottlebrush buck-

eye bloom, followed by evening primrose and passionflower in summer.

Called Bartram's Garden, it is the oldest surviving botanical garden in the United States, and was created in 1728 by John Bartram, a Quaker farmer and self-taught botanist who combed the eastern United States collecting plants and seeds. For a century, well-to-do Philadelphians traveled to Bartram's Garden to buy orchids, peonies, and gardenias for their greenhouses and gardens, as well as cut blossoms for their homes. Bar-

tram and his family operated the garden for decades and—without necessarily knowing it—spared thousands of years of history from being overrun by a spreading metropolis.

The 45-acre site, which is a National Historic Landmark as well as a park operated by the city and the nonprofit John Bartram Association, has had just a few owners—a French Huguenot, three generations of Bartrams, a railroad baron, and Philadelphia Parks and Recreation. It is as rich in history as it is in natural and landscaped

beauty. Recently, in the depths of 2014's cold, snowy winter, a crew of archaeologists from URS Corporation, a San Francisco-based company, were digging in an area of the site behind the house known as Carr Garden. They also recently completed a dig in a meadow down a winding path at the south end of the property.

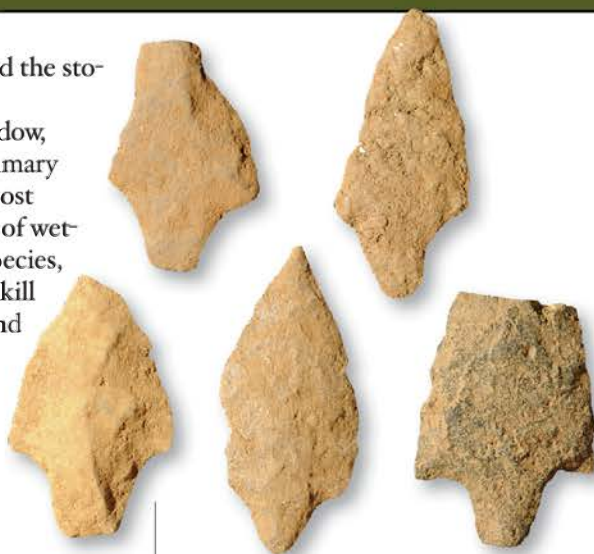
Modern Philadelphia has grown to nearly engulf this stretch of the Schuylkill River Valley. On three sides, Bartram's Garden is hemmed in by industrial properties, public housing, and a railroad line, while across the river is a petroleum refinery and oil storage depot. But this piece of land has survived in part because of a history of caretakers who built lives here and preserved the land, and the record of its past, for future generations of stewards.

Joel Fry, the garden's long-tenured curator and an expert on all things Bartram, had no hint that the land would reveal such a rich record of habitation—one stretching back thousands of years. "There is nothing like [what we discovered] anywhere else in Philadelphia," says Matthew Harris, URS senior archaeologist and principal investigator, who has spent years excavating in urban Philadelphia and the Schuylkill River Valley,

"for density of artifacts and the stories they tell."

At the edge of the meadow, far from the house and primary gardens, at the southernmost property line, was a patch of wetland choked by invasive species, its flow to the tidal Schuylkill River blocked. This wetland was the reason the URS archaeologists dug in the meadow. Two miles downriver, the Philadelphia International Airport needed to expand a taxiway, and, to meet environmental requirements, it was required to compensate for the wetland it was consuming by rehabilitating another, according to Patricia Ann Quigley, an environmental consultant for the airport. The wetland enhancement plans originally included flooding part of the meadow, which required an archaeological assessment before the work could proceed.

In 2012, the URS team opened 74 square-meter excavation units and dug one trench, all in less than an acre of the meadow. The site they uncovered was uncommonly deep. "We found 17,233 artifacts—new prehistoric information not available at other sites—going back maybe



Thousands of prehistoric artifacts, dating to between 2500 and 1600 B.C., emerged from a meadow at Bartram's Garden.

5,000 years, and all the way up to deposits that tell us about the early Philadelphia park system, aspects of women's and children's lives, and the class divide," says Harris. "We could see the artifacts in relation to each other, as they have been for centuries, undisturbed because the ground was never plowed under by agriculture."

The deepest layers they uncovered turned up the biggest surprise: hearths, fire pits, and storage pits, situated throughout the meadow, up to a few feet from the water's edge. They also found that the land had preserved the biggest assemblage of Native American artifacts ever found in Philadelphia. The ceramics they uncovered appear to come primarily from two periods: around 2500 B.C. and from A.D. 800 to 1200. Many of the early rim sherds they found bear "pseudo-cord" designs, which are made by twine impressions. "These are predecessors to more complex and refined geometric designs," says Harris.

Prior to these finds, there had been no surviving record of a Native American presence here, though the Lenape were known to have had villages at Passyunk across the river at one time, and called the Bartram site "Aronaneck." Harris and Fry believe



Portions of several structures, including this 19th-century wall from a greenhouse and engine house, were found in a meadow next to the Schuylkill River.

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A plate decorated with sgraffito may have been a prized possession of French Huguenot Andreas Souplis, who owned a portion of the land in the late 17th and early 18th centuries.

the finds, including expediently made stone tools, are evidence of centuries of seasonal use by hunter-gatherers. "Wild rice grew here," says Harris. "They could have returned regularly for shad and other fish on spawning runs from the ocean, although we don't have net and spear artifacts." The investigators also did not find burials, postholes from a longhouse, or any other signs of permanent settlement.

It is not yet clear who from the tangled history of eastern Native American tribes were using the site. Lenape? Iroquois? Algonquin? "Debate goes on about who was where when," says Fry. "But at [the time of European] contact some Lenape, Iroquois, and Algonquin had already intermingled. We do know that about a mile south of here the Swedes traded with Iroquois."

When Fry mentions the "Swedes," he is referring to a group of seventeenth-century colonists from Sweden, Finland, Germany, and elsewhere who collectively formed a short-lived colony called New Sweden (1638–1655). The Bartram's Garden site lies on the northern edge of the colonial land, and from that group came the land's next occupant: Andreas Souplis, a peripatetic Frenchman who settled down to farm and live there. Souplis, born in Alsace, was a Huguenot, one of a large group of French Protestants who were persecuted for their religious beliefs and driven from France in the seventeenth and eighteenth



centuries. Souplis moved to northern Germany and then sailed to the nascent Pennsylvania colony, where he helped settle Germantown and even became the town sheriff. In his very old age, he moved with his family to Aronaneck. The excavations at the site revealed the finery that Souplis brought with him. "We found ceramics, Staffordshire pottery, smoking pipes, a German Rhine Valley tankard top, a brass furniture pull," says Harris. "High-quality stuff." All the pieces date to between 1680 and 1720, which coincides with the time that Souplis lived there. "The coolest find was a redware piece, about 45 percent complete, in hand-executed sgraffito," says Harris. "The only thing similar that I know of was found at Jamestown." Sgraffito, a technique also used in illuminated manuscripts, involves layering one color over another and then scratching through the upper layer to create a decorative pattern. "It was clear that this was not a common dish, but more likely a cared-for and important possession," Harris adds.

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In 1850, locomotive designer Andrew Eastwick built this mansion, called Bartram Hall, which was demolished in 1896. Broken flowerpots were found at the site of his greenhouse, down the hill from the house.

says Fry. For example, any buildings constructed during Souplis' tenure have not yet been located, and may lie underneath later structures. It would have been common to place new buildings on the sites, even on the exact footprints, of older ones. There were a few nineteenth-century structures on the property, and debris from them might conceal more Souplis material.

Some of what we know of Souplis comes from the deed that transferred ownership of his land from his heirs to John Bartram, the man who gave the site both its name and its place in the history of American botany. Bartram's explorations led him from Lake Ontario to Florida. He wrote books, sold specimens abroad, was chummy with Benjamin Franklin, was named King's Botanist for North America by George III, and is still known as the father of American botany. His son William carried on this tradition, as did William's niece Ann Bartram Carr and her husband Robert. All three generations tended and sold plants on the family property until 1850.

The Bartram house still stands

and is open for guided tours. A greenhouse and barn remain as well. The meadow where so much Native American material was found did not provide any specific evidence of the Bartram ownership of the land. "We do know that John Bartram used this area for grazing his cattle," says Harris. Bartram apparently did not grow crops on this spot. "It was never plowed under for agriculture, so that kept the artifacts in place." The land also rarely floods, further preserving the archaeology below.

However, the preservation of the land itself and the use of acreage closer to the house as botanical gardens are certainly the results of the Bartrams' presence there, and are evidence of their role in the land's stewardship. Ann Bartram Carr made an area close to the house, Carr Garden, her own showplace of exotic species, with 300 varieties of dahlias

around boxwood-lined pathways. In early 2014, the URS archaeologists opened 13 units there, a total of 200 square feet, to help the John Bartram Association in its plans to reconstruct the garden with as much historical accuracy as possible. Extant plans and some early photographs helped guide the archaeologists and showed several paths through the garden, leading to the house and a nearby orchard. "We found the path in five different units," says Harris. "We confirmed its presence, depth, arc, and construc-



tion material. And we found planting beds associated with the path in three units."

Among other surprises—including a number of prehistoric artifacts up to 4,500 years old—Harris found pre-Carr-period soils, likely from the time of her uncle or grandfather. "The key information for us from the archaeology will be placement and path alignment," says Brad Thornton, an associate with LRSLA Studio, the landscape architecture firm at work on the project. "To be practical, for durability we will have to use new materials [on path surfaces] and re-create a look that is historically authentic." Rounding up the historical plants, however, will be difficult because so many have fallen out of fashion.

Indeed, changing fashion in the gardening world may have been the

(continued on page 61)

How a Chicago Doctor Shook Up the Hearing Aid Industry with his Newest Invention

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Reported by J. Page

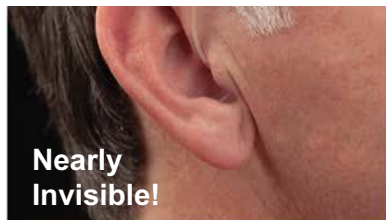
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(continued from page 58)

reason that the Bartram family sold the property in 1850 to Andrew Eastwick, who clearly saw the value in the Bartrams' use of the land as a botanical treasury. Eastwick was born in Philadelphia (a nearby neighborhood is named after him) but made his fortune building locomotives in St. Petersburg, Russia. He retired to Bartram's Garden and built an Italianate mansion atop a hill overlooking the Schuylkill, leaving the Bartram house and garden intact. Evidence of Eastwick's time on the property appeared in the meadow, where the archaeologists located and examined the foundations of his greenhouse—full of broken flowerpots—and another building next door. According to records, this was an engine house, which, in retrospect, provides a connection between the greenery of the park and garden and the industrial sites that line the rest of the Schuylkill. "These buildings were side by side. And one had a floor to support something big," says Harris. "Was it some kind of workshop where he developed his engineering ideas? I'm in love with the idea of finding his St. Petersburg prototype, but it was probably how he heated his house, with a duct running in between. We did find one-inch and three-inch pipes coming from the engine room. Still, that's cool, from a technological standpoint. Who has a tunnel leading to his house for heat?"

Consciously or not, Souplis, Eastwick, and Bartram were extraordinary stewards of this land, even as urban Philadelphia approached. Following Eastwick's death in 1879, industry and urban expansion threatened to finally overwhelm Bartram's Garden. A

A pocketknife was among finds dating to the early 20th century, after Bartram's Garden had been converted to a public park.



Excavations in a riverside meadow (top) prompted changes to the rehabilitation of wetlands along the Schuylkill River (above) in order to preserve cultural deposits. The Philadelphia skyline can be seen from the shore.

campaign to preserve it—in the spirit of Bartram and his gardening traditions—was undertaken and ownership of the site was granted to the city in 1891. With that, the role of steward passed to Philadelphia Parks and Recreation, which created a public park on the property.

Eastwick's mansion burned down in 1896, so the city razed it and filled the site in. They put an ice rink where the wetlands are today, and by the turn of the twentieth century, streetcars ran to the neighborhood, called Kingsessing (another Lenape word). Upper-middle-class Philadelphians, especially mothers and children, socialized and engaged in genteel pastimes, and evidence of this period surfaced in

the excavations of the meadow, which would have been next to the skating rink. "We found lots of tea cups and service

sets, jewelry, ladies' brooches,

and child-size rings and bracelets. And a fancy pocketknife, jacks, marbles, and lots of toys," says Harris. In one cluster, the archaeologists found the remains of a picnic—a tureen, a mustard jar, and an olive container.

Other artifacts from this period suggest a different side of early-twentieth-century life. In one area, out of sight of the main house, the excava-

tors uncovered a large number of milk bottles. Fry and Harris theorize these may have come from a Hooverville, one of the emblematic shantytowns of the Great Depression, not unlike the one that appeared in the middle of New York's Central Park. In a twist, the milk bottles came from Suplee-Wills-Jones dairy—and the Suplees are the descendants of Souplis, the original landowner.

In light of all the finds made in the meadow and around the property, engineers from URS curtailed and rejiggered the wetlands mitigation project so that the meadow has not been flooded. And this is what preservation today requires, something that Bartram and the other owners probably never would have thought to consider: accounting for the site's natural history as well as its human history. ■

Writer Margaret Shakespeare lives in New York City and Long Island, and comes from generations of proud Philadelphians.

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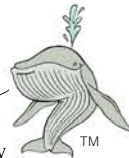
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AIA Supports Preservation of Ancient Maya Murals in Belize



sealed off to protect it and its contents.

In addition to these conservation measures, MRP will establish a permanent outreach program for the local community, working with local school groups, church groups, landowners, and others to promote research and preservation at the site.

THE ARCHAEOLOGICAL Institute of America (AIA) awarded its most recent Site Preservation Grant to a conservation and outreach project at the Early Classic Maya site of Tulix Mul in Belize. The project, directed by Thomas Guderjan of the Maya Research Program (MRP), will use the funds to record and protect a rare and fragile mural found at the site and to establish outreach programs to the local community.

Tulix Mul is located in the middle of a working cattle ranch in northwestern Belize and was discovered by MRP in the course of their decades-long investigation of the region. The site was established around A.D. 300 and the mural was painted soon after. The mural is located on the back wall of a room with a vaulted ceiling in what Guderjan believes was the house of a locally important lineage head. The mural, the subject of which is still unclear, was covered with plaster in ancient times and the room was intentionally filled in around A.D. 450. The site was abandoned around A.D. 850. The artwork was discovered during the excavation of the room because small portions of the ancient plaster had peeled off the wall, partially exposing

the painting underneath.

While Tulix Mul is currently protected by a fence and local farm workers assist with security, the mural is threatened by exposure, looting, and damage at the hands of visitors unaware of or indifferent to the fragile nature of the painting. MRP is planning a multifaceted, long-term program to protect both the mural and the site. The first step will be to curb the physical destruction of the site and the artwork, and MRP is in active negotiations with the landowner to purchase the site. Then, under the supervision of a professional conservator, MRP will remove the plaster layer and fully expose the mural. Once the painting is recorded digitally and manually, the entire room will be

Public programs will include lectures, community workshops, a tour guide training program, and open lab days. Through these efforts, MRP hopes to increase local awareness of the significance and fragility of the archaeological remains at Tulix Mul and provide opportunities for the local community to derive some economic benefits from the site.

MRP's holistic approach to site preservation is consistent with the AIA Site Preservation Program's mission and a good example of the type of innovative and inclusive projects that the Program supports. To read more about our projects, visit archaeological.org/sitepreservation. The application deadline for the next round of grants is October 15, 2014.



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Best Practices in Site Preservation Award Presented to Society for California Archaeology Program

THE AIA CONSERVATION AND Site Preservation Committee presented the 2014 Best Practices in Site Preservation Award to the California Archaeological Site Stewardship Program (CASSP) for its efforts to involve and train local volunteers in the preservation of archaeological sites in California.

Tens of thousands of archaeological sites located on public lands in California are protected by federal agencies, such as the National Park Service and Bureau of Land Management, and local state agencies. These agencies use a combination of active monitoring, fencing and signage, interpretive trails, and public outreach to safeguard the sites. But these approaches do not involve active public participation.

In 1999, the nonprofit Society for California Archaeology (SCA) established CASSP, a statewide program that trains local volunteers to act as site stewards, whose job it is to visit



assigned sites on public lands and report their conditions to the local supervising archaeologist. By regularly monitoring sites, site stewards ensure that potential problems are detected early and corrected quickly.

Since the program's inception, nearly 1,400 people have participated in 75 training workshops across the state. Site stewards agree to make one site visit a month, and each year, more than 300 volunteers monitor more than 400 sites. By involving the public in the protection of sites, CASSP is breaking down the barriers between archaeologists and the public,



encouraging a better understanding of archaeological heritage, increasing the visibility of the past, and ensuring the protection of archaeological sites.

The Best Practices Award, established by the AIA Conservation and Site Preservation Committee in 2011, seeks to identify and promote best practices in the interdisciplinary field of site preservation. The award, which includes a cash prize of \$5,000 intended to further the work of the recipient, is presented annually each year at the AIA's Annual Meeting in January. Nominations for the next award are due May 1, 2014.

Staffordshire Hoard Conservation Project Receives AIA Award



AT THE AIA ANNUAL MEETING in Chicago, the Conservation and Site Preservation Committee presented the 2014 Conservation and Heritage Management Award to the Staffordshire Hoard Conservation Project (SHCP). Since its inception in 2010, SHCP has used an innovative, team-based approach to the conservation of one of the largest Anglo-Saxon hoards ever found.

The Staffordshire Hoard, discovered

in 2009, is a collection of approximately 3,500 gold and silver objects that was found in a field in the village of Hammerwich, in Staffordshire. The spectacular find made headlines in the United Kingdom and around the world, and public demand to view the objects was so great that select pieces were displayed at the Birmingham Museum Art Gallery (BMAG). In just 19 days, more than 42,000 people visited the display. The hoard is now owned by the BMAG and the Stoke

Potteries Museum and Art Gallery.

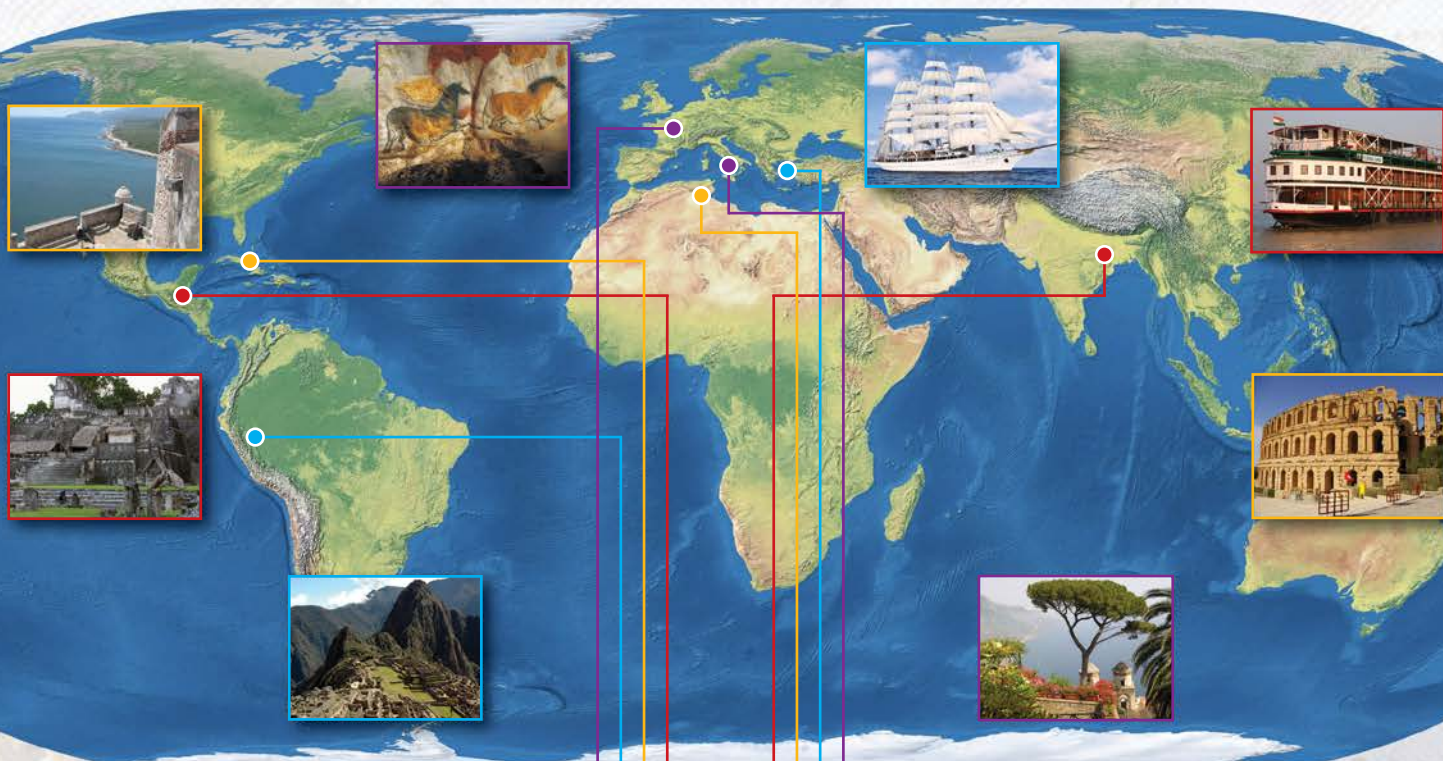
Because of the public interest in the hoard, SHCP developed an open, collegial, and team-based approach to the conservation of the objects that attracted several experienced professional conservators. Participation in the project by these experts ensured that the highest standards of conservation were met. Members of the public were involved in the process through lectures, studio tours, family days, and written and video blogs created by the conservation team. Through these efforts, SHCP created a vibrant community of professionals and non-professionals that is involved in the project and invested in the long-term preservation and presentation of the objects.

The AIA Conservation and Heritage Management Award is presented each year at the AIA Annual Meeting in January. Nominations for the next award are due May 1, 2014.

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It was a dangerous year for the inhabitants of Sardis. In A.D. 17 an earthquake, which the ancient historian Pliny called “the greatest earthquake in human memory,” destroyed the great city, once the capital of the Lydian Empire and, at the time, a Roman imperial provincial capital. It took the Sardians decades to rebuild—but they weren’t taking any chances. Members of Harvard University’s Sardis project recently discovered a simple, locally produced bowl, with a similar bowl used as a cover, that had been buried under the floor of a domestic or public space sometime after A.D. 65. Inside the bowl was a collection of artifacts that project archaeologist Elizabeth DeRidder Raubolt of the University of Missouri describes as having been used for rituals of “protective magic.” These included an eggshell, which she describes as “about the size of medium organic egg, not the jumbo or extra large ones,” as well as a bronze needle perhaps used for piercing the eggshell, a pebble, iron implements, a bronze nail, and a bronze coin.

As is perhaps appropriate in the city that Herodotus tells us was the first to mint coinage in the seventh century B.C., it is the coin that provides the most compelling evidence of the Sardians’ desire to avoid another disastrous event. The bronze coin, which dates to between A.D. 62 and 65, depicts the emperor Nero on the obverse, and, on the reverse, an image of the god Zeus Lydios, a native deity of storms and thunder. “Clearly the Sardians chose this coin specifically as a plea for there to be no more earthquakes,” says DeRidder Raubolt.

WHAT IS IT*Ritual deposit***CULTURE***Roman***DATE***Third quarter of the 1st to early 2nd century A.D.***MATERIALS***Pottery, eggshell, bronze, iron, stone***FOUND***Sardis, western Turkey***DIMENSIONS***Bowl: about 5 inches wide, 2 inches tall*

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